Lecture 1

Introduction to Machine Learning

Welcome to CS189/289

EECS 189/289, Spring 2026 @ UC Berkeley
Jennifer Listgarten and Alex Dimakis
All emails should go to: cs189-instructors@berkeley.edu

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Roadmap

- Introductions
- CS 189/289 Goals and Plans
- What Is Machine Learning?
  - Definition
  - History
- The ML Process (Lifecycle)
- Logistics

Introductions

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Jennifer Listgarten

- **Professor** at EECS UC Berkeley (2017-now).
  - Steering committee member of BAIR,
  - Center for Computational Biology
  - UC Berkeley Bioengineering program
- **Research:** ML and Molecular Biology
  - How ML can be used to advance protein engineering
  - Molecule design for drug discovery
- **I also teach**: CS294 (ML+Statistics+Biology)
- This semester, I'm:
  - Teaching this course, advising my students, and doing ML+Bio research

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Alex Dimakis

- **Professor** at EECS UC Berkeley (2025-now).
  - Sky Computing Lab and a member of BAIR
- **Research:** Generative AI,
  How to train agents, how to create datasets.
- E.g. Openthoughts-Agent

- **I also teach**: EE120 (Signals and Systems)

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CS 189/289 Goals and Plans

- Introductions
- CS 189/289 Goals and Plans
- What Is Machine Learning?
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Why Are We Excited to Teach CS189?

- **The field shifted under our feet** → we're redesigning now so we **surf, not chase**.
- **We listened to student pain points** → we're tightening the storyline to **math** → **code** → **experiments**, with unified notation and right-sized and reasonable prerequisites.
- **Classroom work should feed research** → projects will connect to live problems so **students** → **collaborators**.
- **Berkeley should set the bar** → we aim to produce a shareable resource that others can adopt and adapt.
- We are trying **MANY NEW THINGS** some things will break… but we hope you will enjoy this class.

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By the End of CS189, You Should:

- Have a **rigorous foundation** in core ML concepts and algorithms, connecting the math to the methods we use.
- Be able to **implement, train, and debug** standard models in Python/PyTorch and move from data prep to experiments.
- Be able to **design sound evaluations** and document your work so results are **reproducible and responsible**.
- Be **prepared for advanced courses and research**.

These are ambitious goals!

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Accel in Advanced ML Courses

Advanced ML course:

- Computer vision (180/280), NLP (288), RL (285), Statistical Learning Theory (281), and Many Special Topics Courses on AI (194/294)

Require understanding of basic concepts:

- Model design
- Stochastic Gradient Descent
- Maximum Likelihood
- Loss Structure
- Supervised Learning
- Unsupervised Learning
- Classification
- Regression
- Regularization and the Bias/Variance Tradeoff
- Dimensionality Reduction
- Clustering and Density Estimation
- Experiment Design and Tracking

…

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What Is Machine Learning?

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What Is Machine Learning (ML)?

Software systems that improve (learn) through data.

Classic Example: What is Spam?
- Difficult to define
- Easy to demonstrate

Classic Example: Face detection?
- Difficult to program
- Easy to demonstrate

<u>Basic Recipe:</u>

- Use examples (**data**) to teach (**fit**) a **model**.
- Use the **model** to **make decisions**.

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What Is **Artificial Intelligence?**

Artificial intelligence (AI) refers to the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

-- Wikipedia (2025)

The primary technology behind modern AI is

Machine Learning (ML)

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Artificial Intelligence Is the **Goal**
Machine Learning Is the **Method**

Terms are often used interchangeably (even by experts).

An entrepreneur's note on AI marketing

If you are selling

- **Machine Learning** then you call it **Deep Learning.** ***(2015-2020)***
- **Deep Learning** then you call it **AI**. ***(2020-2024)***
- **AI** you call it an **Agent**. ***(2024-Now)***

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You have (probably) already done **Machine Learning**

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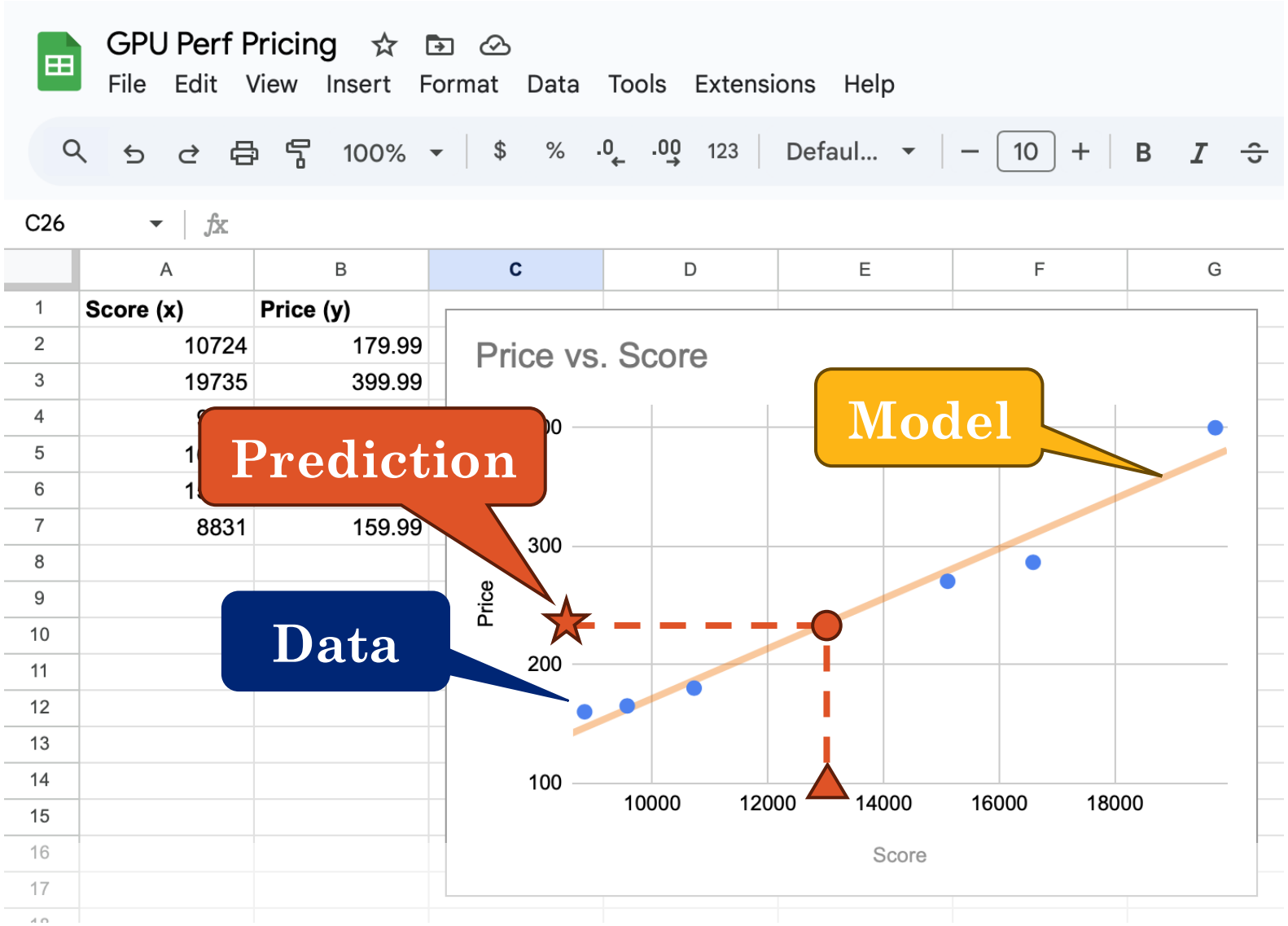
Linear Regression is Machine Learning



<u>Basic Recipe:</u>

Training: Use **data** to teach (fit) a **model**.

Inference: Use the **model** to make **predictions** (decisions).

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Overview of the landscape

Terms used:

Data science
Machine Learning
Artificial Intelligence
Big Data
Data Mining
Statistics

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Terms used:

Data science
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@jeremyjarvis
"A data scientist is a statistician who lives in San Francisco"

@BigDataBorat
Data science is statistics on a Mac.

@josh_wills
Data Scientist (n.): Person who is better at statistics than any software engineer and better at software engineering than any statistician

(anonymous)
The difference between statistics and data science is about 30k per year.

(anonymous)
It is statistics if it's done in R.
It is Machine learning if it's done in Python
It is AI if it's done in Powerpoint.

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Overview of the landscape

Statistics: The very first line of the American Statistical Association's definition of statistics is "**Statistics is the science of learning from data...**" Given that the words "data" and "science" appear in the definition, one might assume that data science is just a rebranding of statistics.
Statisticians are not happy they are not getting the research funding and salary bumps involved.

Data science: Broad and modern term. Includes much more software engineering knowledge. Usually done in Python (as opposed to R/Stata/SAS/SPSS/Excel etc). Includes analytics on bigger datasets (e.g. terabytes or petabytes by using tools like Apache Spark, Hadoop Mapreduce which enable distributed processing. Includes data collection and data cleaning pipelines (data engineering/ data wrangling). Connections to database backends and web-serving front-ends. Includes to some extend machine learning and AI as sub-areas.

Machine learning: The more mathematically complex part of data science focused on modeling (as opposed to software). Includes supervised learning, predictive modeling, unsupervised learning (like clustering), text and image understanding.

Artificial Intelligence: Broad and classic term that includes machine learning as a sub-discipline. *Allowing computers to do things that humans do when they say they are thinking*. Includes perception (image understanding, speech understanding), Language translation, playing games, statistical machine learning techniques but also logic-based symbolic AI, reasoning, planning, and robotics).

Data Mining: Applied version of Machine learning. Includes more large-scale software and performance issues. Also intersects the database community.

Big Data: Focused on scaling data analytics to very large data sets. Part of data science that will hopefully follow the information superhighway and internet surfing to obsolete historical nomenclature.

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In terms of research communities:

Statistics
Research published in stats journals. Top venues: **Annals of Statistics, JASA, Journal of the Royal Statistical Society**.

Data science:
Not a properly defined research community.

Machine learning:
Research published in top ML conferences: **NeurIPS**, **ICML**, also more recently **ICLR**. Also includes **KDD** (more applied, data mining).
https://iclr.cc/virtual_2020/index.html

Artificial Intelligence:
Includes ML conferences but also **AAAI** and **IJCAI** as top venues.

Data Mining: Applied version of Machine learning. Includes more large-scale software and performance issues. Also intersects the database community.
Research published in top Data Mining conferences: **KDD**, **SDM**.

Big Data:
Not a properly defined research community.
Engineers who can setup Hadoop/Spark clusters. Can work on data directly on disk and process at massive scale.

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A taxonomy for machine learning

- We will develop a taxonomy of different machine learning problems.
- 1. Supervised versus Unsupervised.

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Supervised learning: Binary classification

- Given a table of training data containing **features** (x1,x2,..) and a target variable y we want to predict.
- Example: Taste-test: Predict if a new beverage will be evaluated as having 'Great taste' from a focus group.

	Acidity(A)	**Sweetness (S)**	**y='Great taste'?**
Bev1	0.8	0.8	1
Bev2	0.3	0.25	0
Bev3	0.2	0.8	0
Bev4	0.3	0.7	0
Bev5	0.9	0.7	1

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Supervised learning: Binary classification Data-driven Taste Test example

- Jargon: Every row is called a training set sample.
- **We have 2 features** (Acidity and Sweetness). In practice we may have many more (Color, Carbonation level, ..). In statistics these are also called **covariates**, **factors** or **regressors**.
  (in some stats literature, a factor is a categorical feature, a covariate is a continuous feature)

The **target variable y** is now binary (good taste or not): **Binary classification**.

y **could have multiple levels**: (Poor, mediocre, Good, Great): **Multi-label classification**

Or y could be a **continuous** number to predict (Taste score from 1:100): **Regression.**

	Acidity(A)	**Sweetness (S)**	**y='Great taste'?**
Bev1	0.8	0.8	1
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Supervised learning: Binary classification

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How we do we make a good prediction rule?

A first idea: Lookup Table

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Supervised learning: Binary classification

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev1	0.8	0.8	1
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How we do we make a good prediction rule?

~~A first idea: Lookup Table~~

That won't work. We need some kind of rule. Ideally simple. We now discuss two common frameworks for this: **Decision Trees** & **Linear**

Binary classification with a short tree: A decision stump

	Acidity(A)	Sweetness (S)	y='Great taste'?	Model 1 predicts
Bev1	0.8	0.8	1	
Bev2	0.3	0.25	0	
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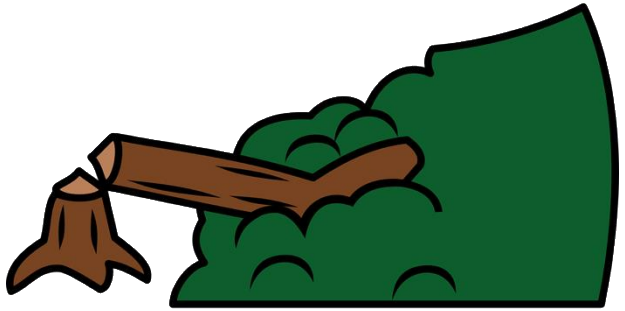

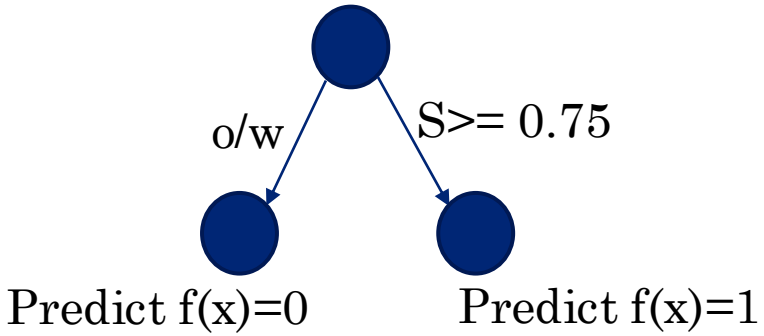

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Binary classification with a short tree: A decision stump

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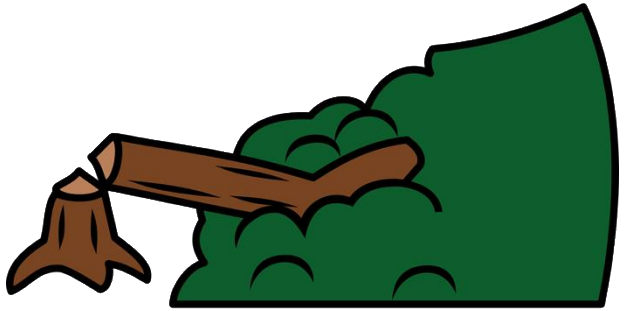
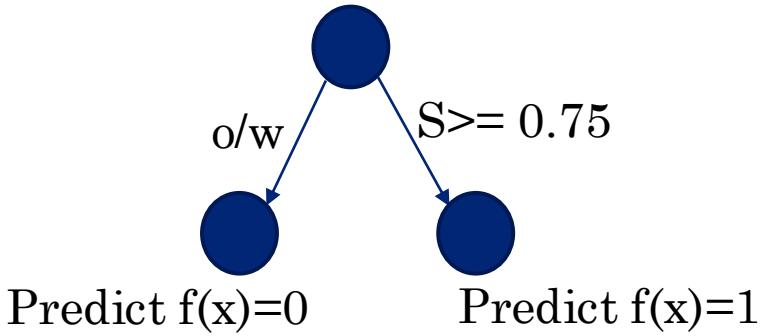
Bev5	0.9	0.7	1	0

Accuracy of this model on the training set is:

?/ 5





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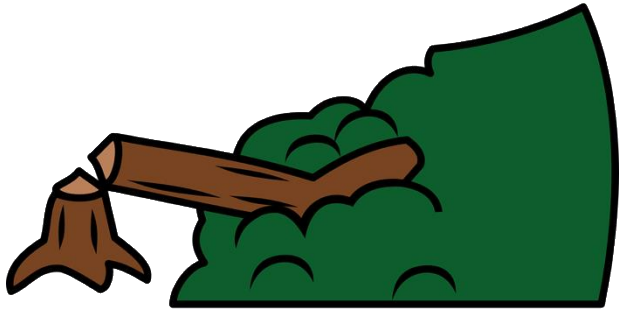
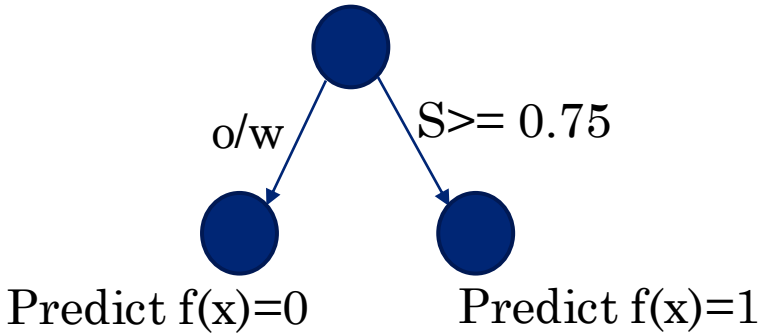
Accuracy of this model on the training set is:

3/ 5





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Partitioning the feature space

	Acidity(A)	Sweetness (S)	y='Great taste'?
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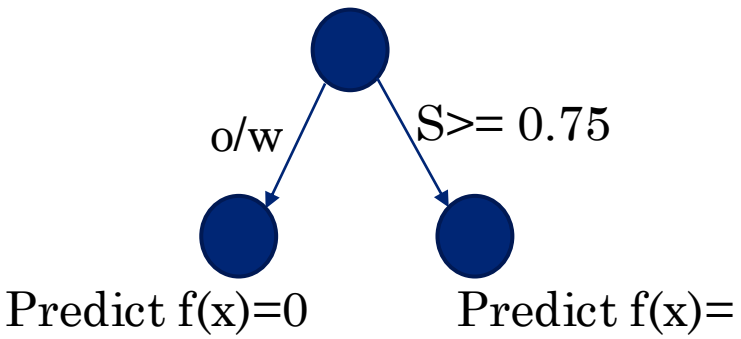

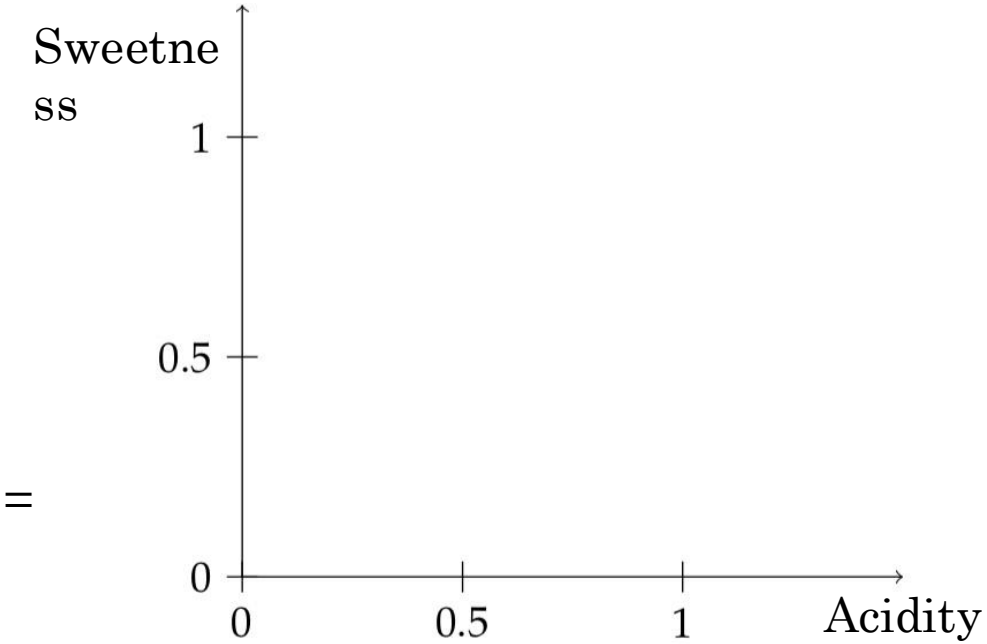

Lets position the data on this feature space

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Partitioning the feature space

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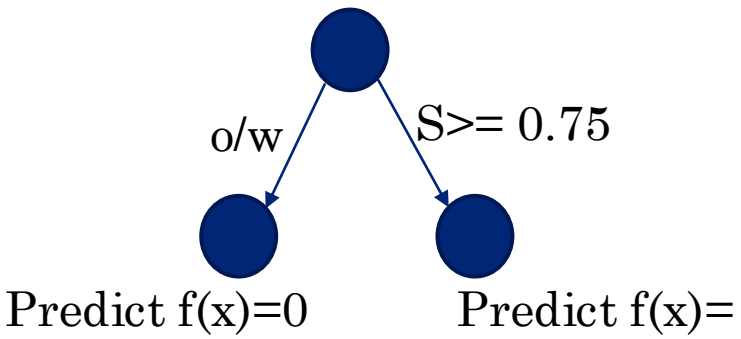

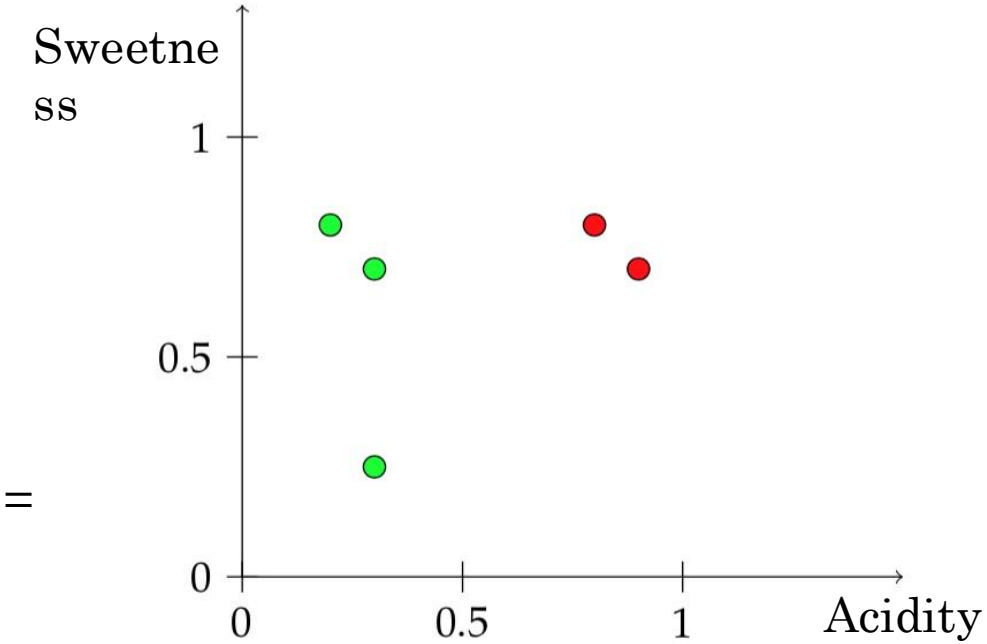

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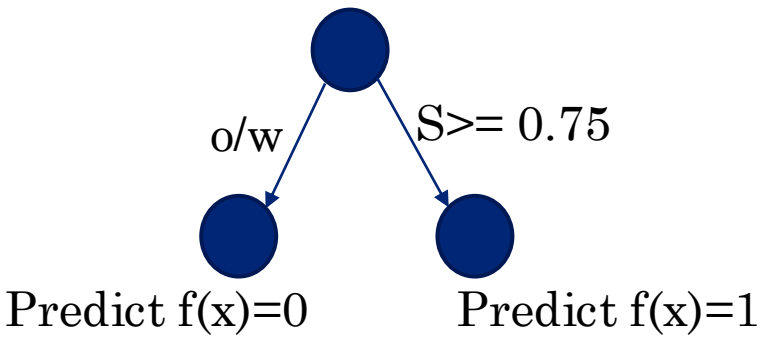

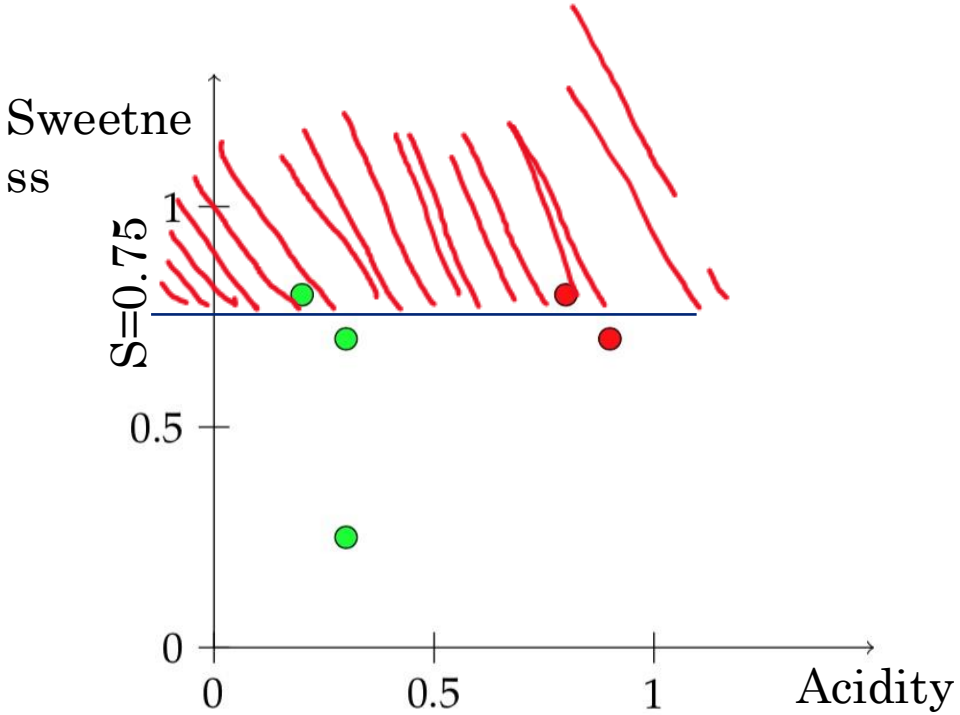

Each binary classifier has a **decision region**: How it partitions the feature space.

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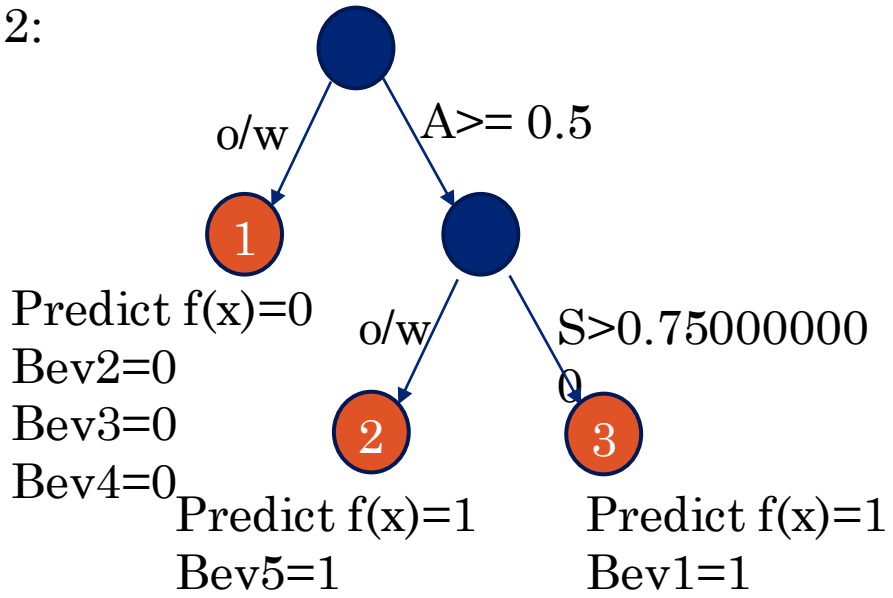
Binary classification with depth 2 decision tree:

	Acidity(A)	Sweetness (S)	y='Great taste'?	Model 2 predicts
Bev1	0.8	0.8	1	**1**
Bev2	0.3	0.25	0	**0**
Bev3	0.2	0.8	0	**0**
Bev4	0.3	0.7	0	**0**
Bev5	0.9	0.7	1	**0**

Accuracy of this model on the training set is:
1/5 = 20% -> 4/5

Loss function we use is the normal 0-1 accuracy.

Model 2:



?

This model **splits first on A with threshold 0.5**
and **then on S with threshold 0.75.**

f(x)= yhat =(ideally)= y.
f(A=0.8,S=0.8)= 1

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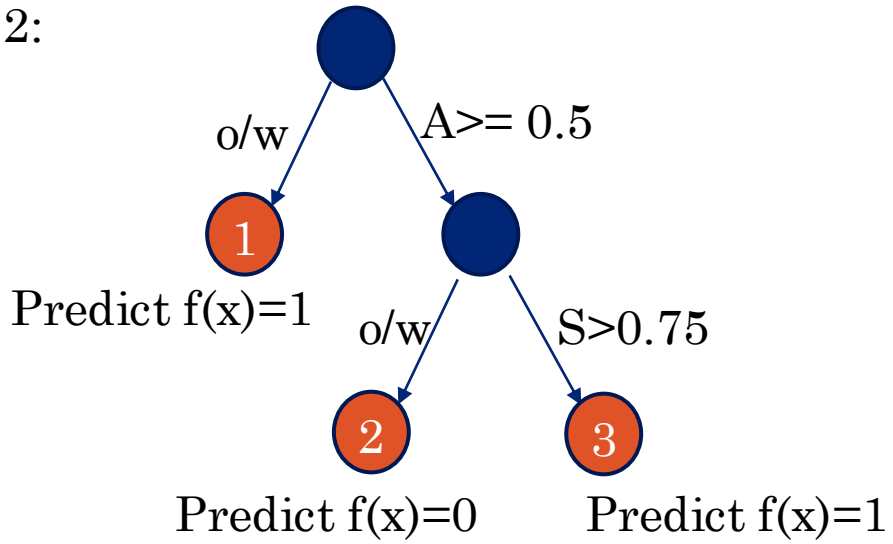
Binary classification with depth 2 decision tree:

	Acidity(A)	Sweetness (S)	y='Great taste'?	Model 2 predicts
Bev1	0.8	0.8	1	1
Bev2	0.3	0.25	0	1
Bev3	0.2	0.8	0	1
Bev4	0.3	0.7	0	1
Bev5	0.9	0.7	1	0

Accuracy of this model on the training set is:

?

Model 2:



This model **splits first on A with threshold 0.5**
and **then on S with threshold 0.75.**

Could you get better training accuracy by labeling leaves differently ? YES.
What is a general algorithm of finding the labels of the leaves that maximize the training accuracy for a given tree ?
Run one epoch of the dataset, place every sample in one leaf, label leaf with majority of training labels.

Binary classification with depth 2 decision tree:

	Acidity(A)	Sweetness (S)	y='Great taste'?	Model 2 predicts
Bev1	0.8	0.8	1	1
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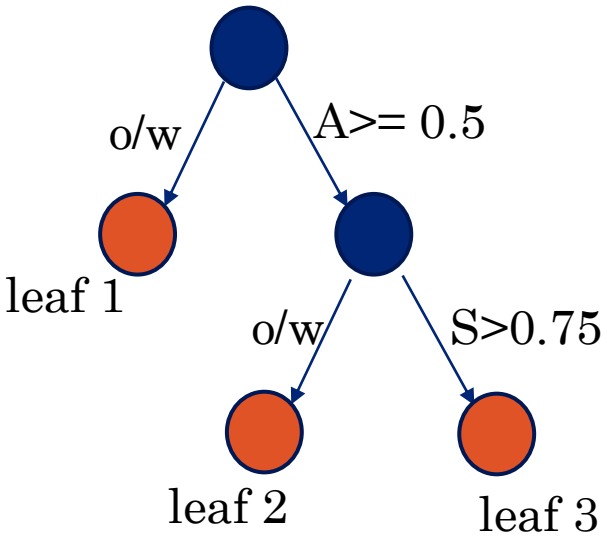
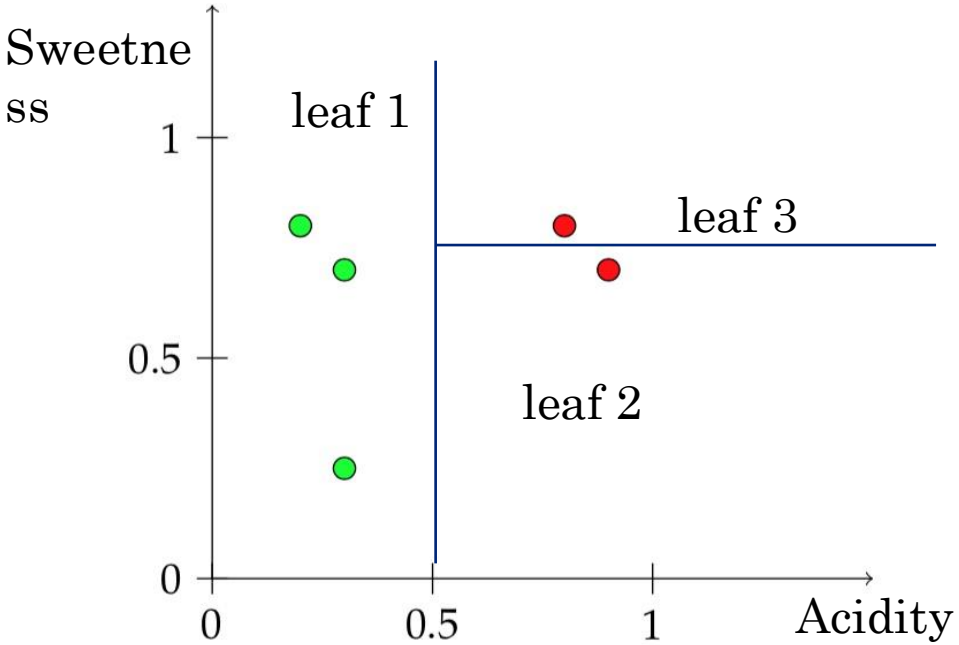
Bev5	0.9	0.7	1	0

Accuracy of this model on the training set is:

Model 2:





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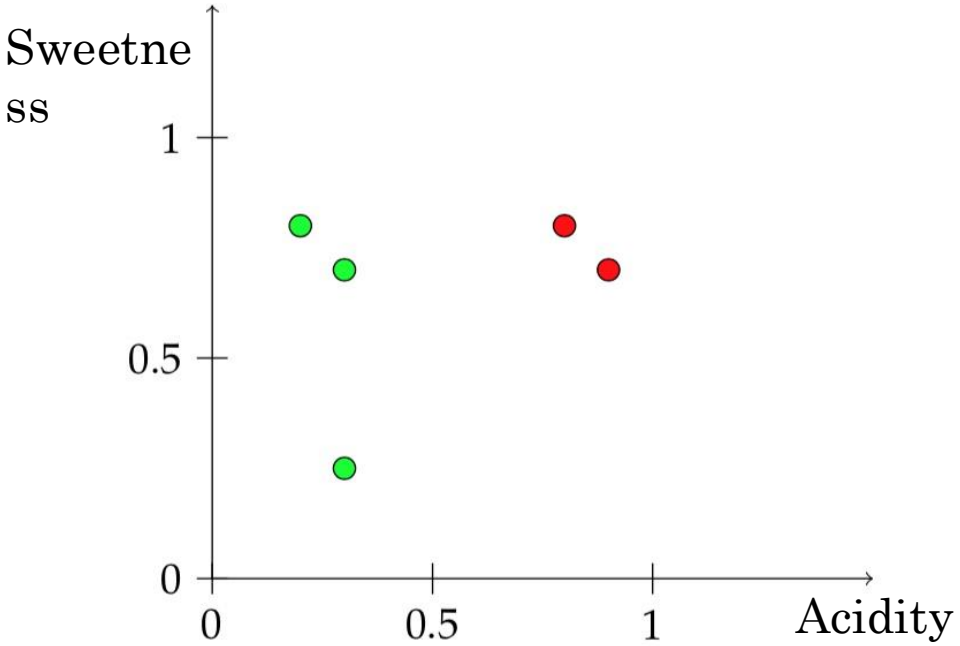
Binary classification with a **linear classifier**

	Acidity(A)	Sweetness (S)	y='Great taste'?	Model 3 predicts
Bev1	0.8	0.8	1	?
Bev2	0.3	0.25	0	?
Bev3	0.2	0.8	0	?
Bev4	0.3	0.7	0	?
Bev5	0.9	0.7	1	?

Model 3:

$f(A,S) = 1$ if $A+S-1 \geq 0$
$\quad\quad\;\; 0$ otherwise

Compute the predictions of this model.
Draw the decision boundary



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Binary classification with a linear classifier

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Bev1	0.8	0.8	1	?
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Bev3	0.2	0.8	0	?
Bev4	0.3	0.7	0	?
Bev5	0.9	0.7	1	?

Model 3:

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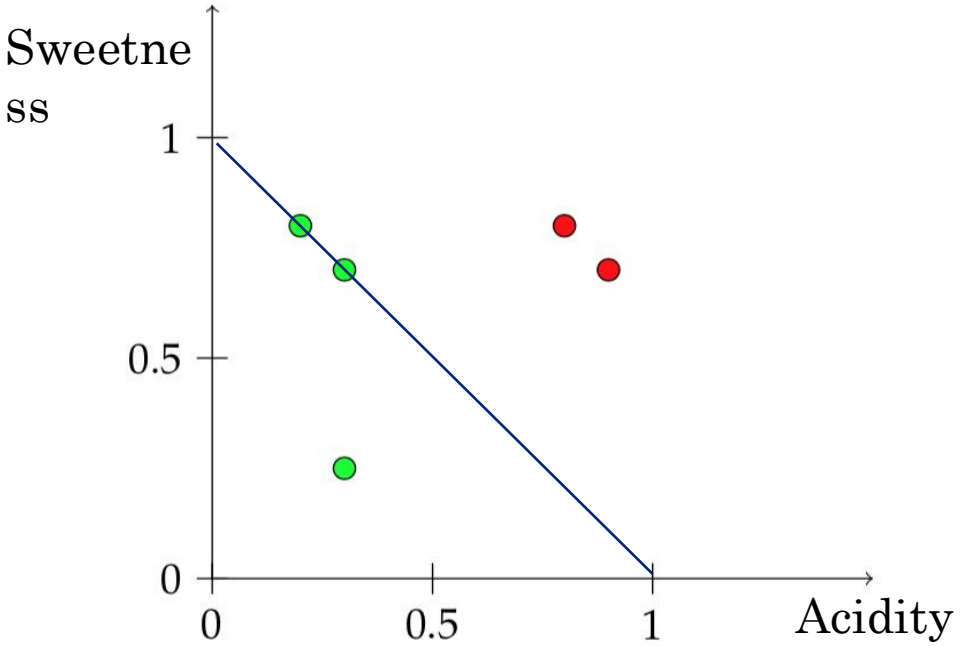

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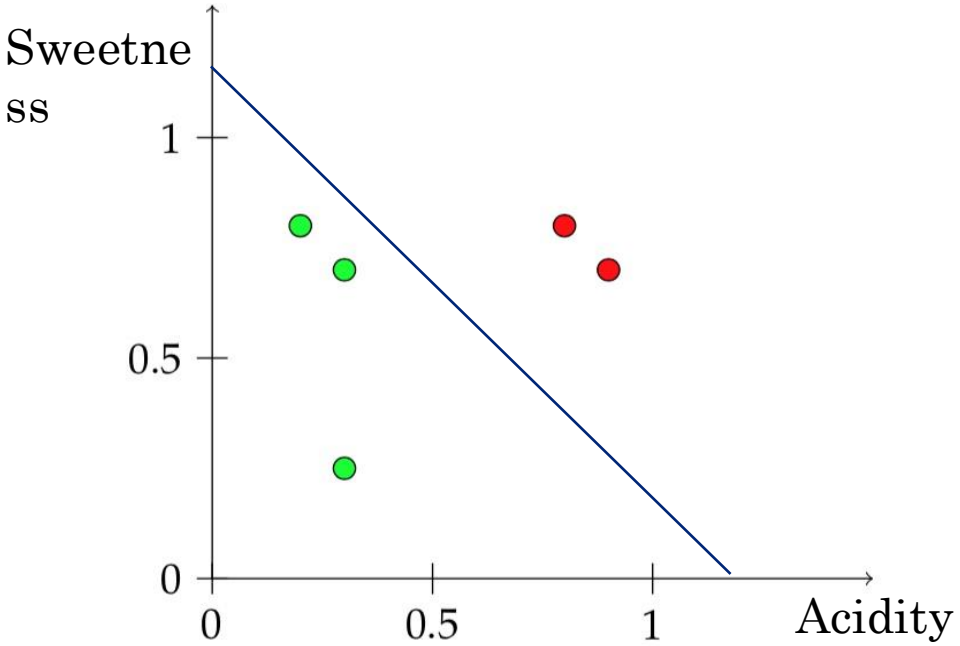
Binary classification with a linear classifier

	Acidity(A)	Sweetness (S)	y='Great taste'?	Model 3 predicts
Bev1	0.8	0.8	1	?
Bev2	0.3	0.25	0	?
Bev3	0.2	0.8	0	?
Bev4	0.3	0.7	0	?
Bev5	0.9	0.7	1	?

Model 4:

$f(A,S) = 1$ if $A+S-1.3 \geq 0$
$\quad\quad\;\; 0$ otherwise

Compute the predictions of this model.
Draw the decision boundary



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Training and Test set

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev1	0.8	0.8	1
Bev2	0.3	0.25	0
Bev3	0.2	0.8	0
Bev4	0.3	0.7	0
Bev5	0.9	0.7	1

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev6	0.7	0.7	?
Bev7	0.1	0.1	?
…	…	…	?
…	…	…	?
…	…	…	?

- The goal of prediction is **to predict the future, not the past**.
- We have a few more beverages with different features and we want to predict if people will like them, without administering a study for each one.

(Could have billions of combinations of features)

How do we build a prediction model that goes from (Acidity, Sweetness)-> will people like it?

How do we evaluate if the model works well. (Model evaluation)

How do we interpret if works well for the right reasons. (Model interpretability)

We will spend a lot of time covering all these topics

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There is always one very bad model

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev1	0.8	0.8	1
Bev2	.3	0.25	0
Bev3	0.2	0.8	0
Bev4	0.3	0.7	0
Bev5	0.9	0.7	1

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev6	0.7	0.7	?
Bev7	0.1	0.1	?
…	…	…	?
…	…	…	?
…	…	…	?

- Claim: For any training set (arbitrarily large) There is a model M_{BAD} (A, S) that has perfect training accuracy and terrible predictive performance. Construct it.

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There is always one very bad model

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev1	0.8	0.8	1
Bev2	.3	0.25	0
Bev3	0.2	0.8	0
Bev4	0.3	0.7	0
Bev5	0.9	0.7	1

	Acidity(A)	Sweetness (S)	y='Great taste'?
Bev6	0.7	0.7	?
Bev7	0.1	0.1	?
…	…	…	?
…	…	…	?
…	…	…	?

- Claim: For any training set (arbitrarily large) There is a model M_{BAD} (A, S) that has perfect training accuracy and terrible predictive performance. Construct it.
- $Model_{Bad}$(Ainp, Sinp): if Ainp,Sinp in training set row i, output yi (true label for row i)
- else: output 0.
- $Model_{bad}$ is **memorizing** the training set but it cannot **generalize**.
- Ideally we want the simplest model that has good accuracy on the training data.
- Occam's razor: The simplest explanation that fits the data is usually the correct one.

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Machine Learning

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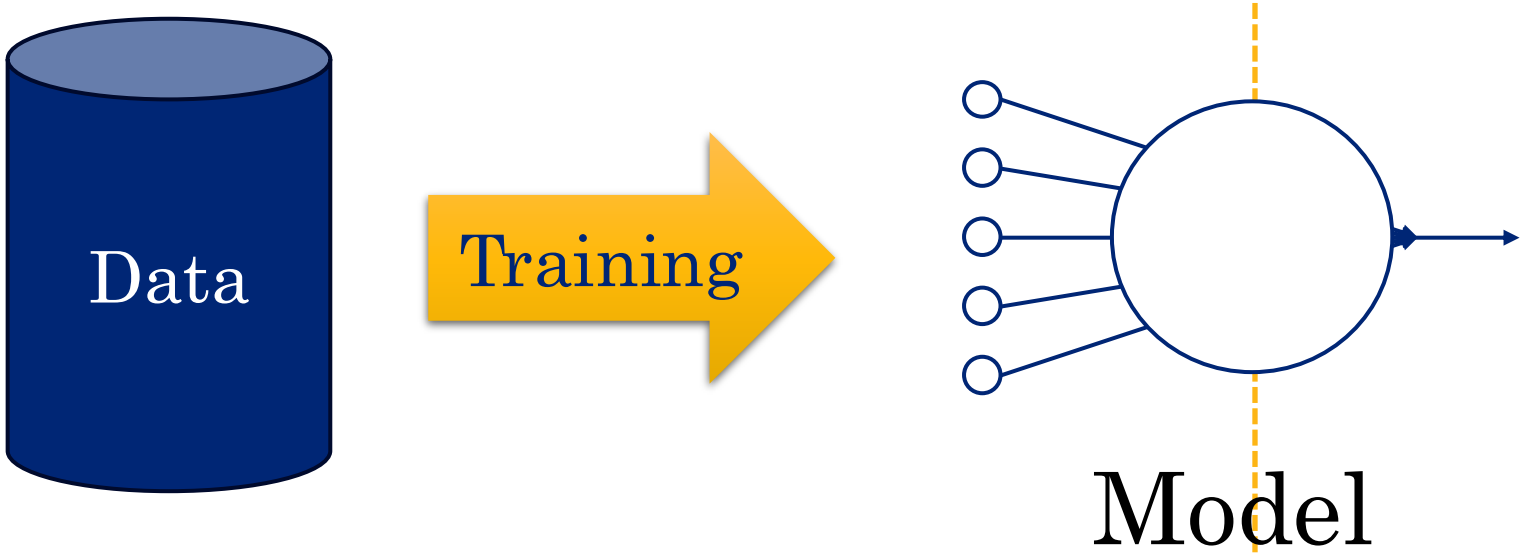
Training



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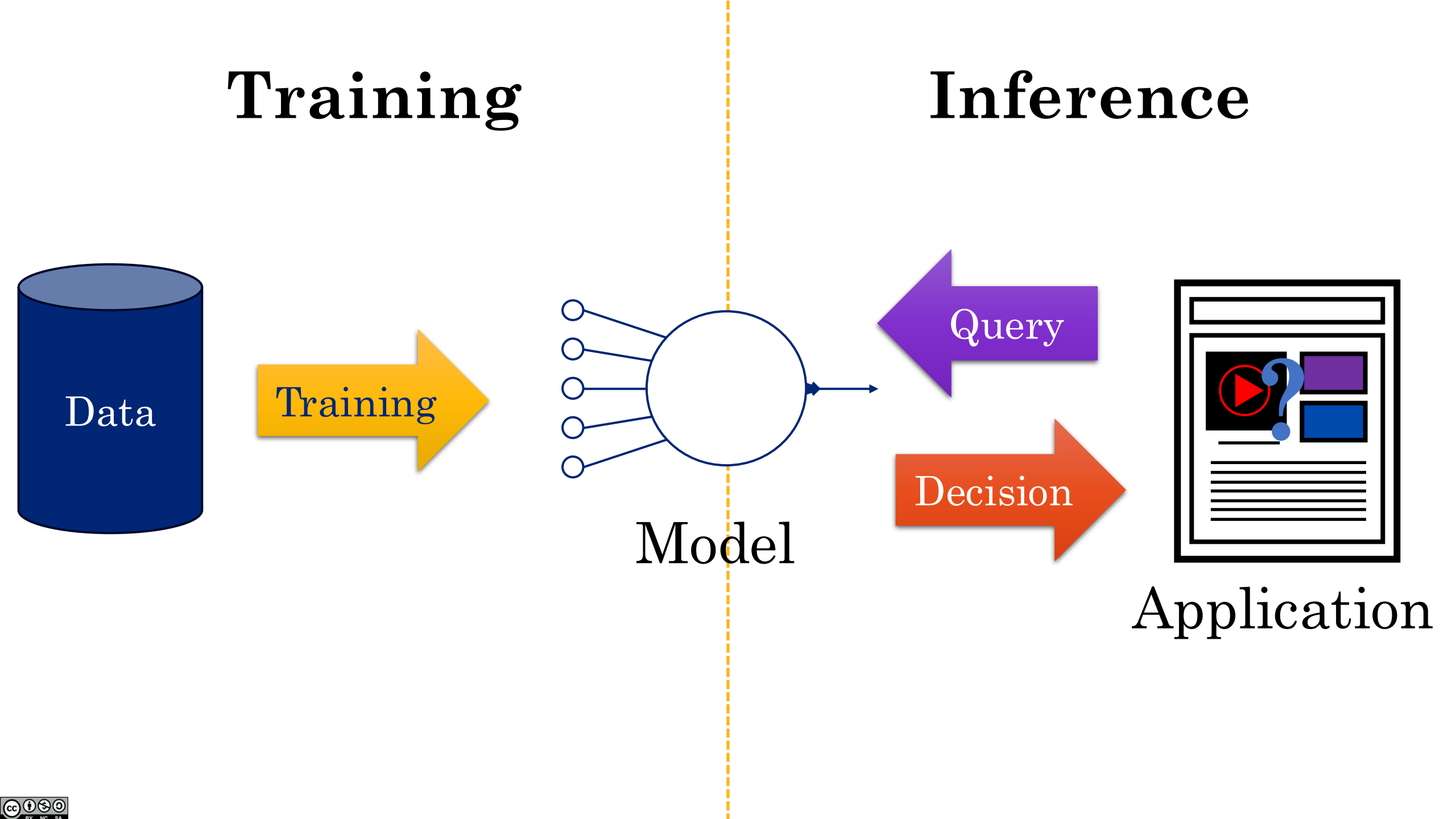
Training
Inference
Data
Training
Model
Query
Decision
Application

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When should I used Machine Learning?

What kinds of problems are **Machine Learning Problems**?

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Kinds of Problems

Engineering Problem: Can be solved with a direct, specifiable algorithm or a set of hand-written rules.

Machine Learning Problem: For which it is easy to demonstrate or evaluate the solution but difficult to directly implement.

A Human Problem: The problem cannot be well specified and/or human judgement is required.

→ Often require ***Engineering + ML + Humans*** 👉

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A Machine Learning Problem

A problem for which it is **easy to demonstrate** or **evaluate the solution** but **difficult to directly implement**.

Machine Learning Solution:

The system **learns the desired behavior** (e.g., prediction, representation, or a policy) through **demonstration** or **experience.**

Example:

Determine if a text message is spam

How do you define Spam?

Spam is **difficult to define** and **depends on the receiver**.

Easier to **demonstrate examples** and **learn a function** to detect

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Is Chatting a Machine Learning Problem?

Example (ChatGPT):

Engage a human in a productive conversation

How do you program this?

```
  Welcome to
                    EEEEEE  LL      IIII   ZZZZZZ   AAAAA
                    EE      LL       II       ZZ   AA   AA
                    EEEEE   LL       II     ZZZ    AAAAAAA
                    EE      LL       II    ZZ      AA   AA
                    EEEEEE  LLLLLL  IIII ZZZZZZ    AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.


ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

ELIZA (1966)
Rule based
Conversational System

https://en.wikipedia.org/wiki/ELIZA

https://web.njit.edu/~ronkowit/eliza.html

Entertaining but **it can't do your homework.**

We can **demonstrate** good conversations.

We can **judge** good conversations.

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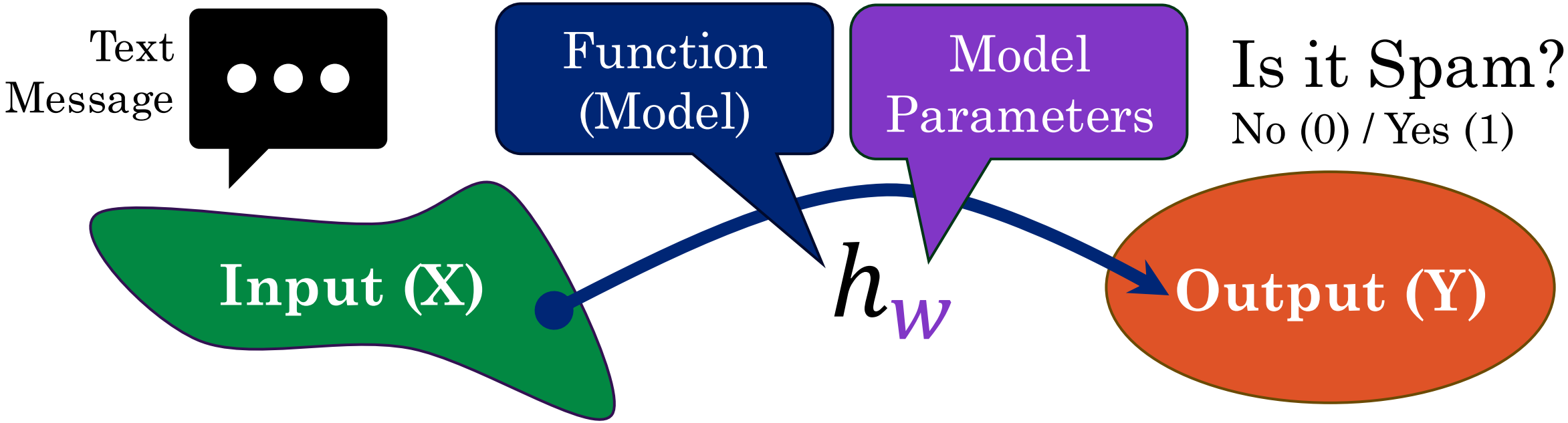
Machine Learning as **Learned Function Approximation**



Machine learning becomes the process of **"learning"** the **model parameters** from data or interaction with the world.

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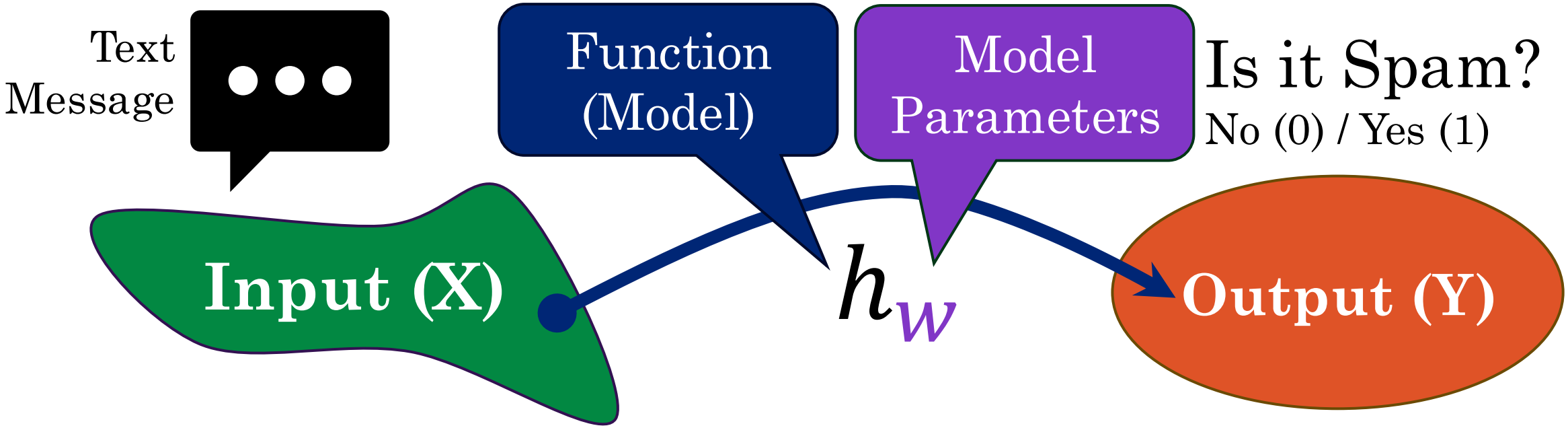

Learning Settings

	Supervised (Demonstration)	Unsupervised	Reinforcement (Reward)
Observe:	**{(X,Y)}**	**{X}**	**X, reward(.)**

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Supervised Learning

Trying to **learn a relationship** between **observed {(X,Y)} pairs**.

Classification

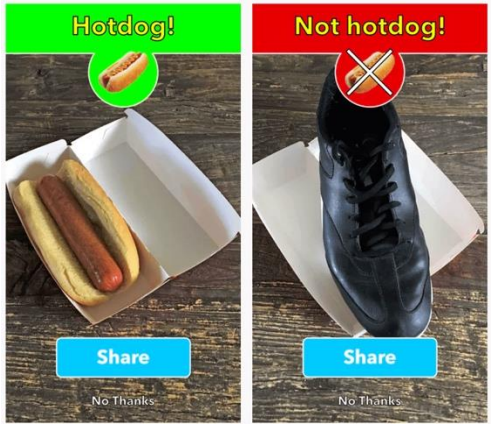


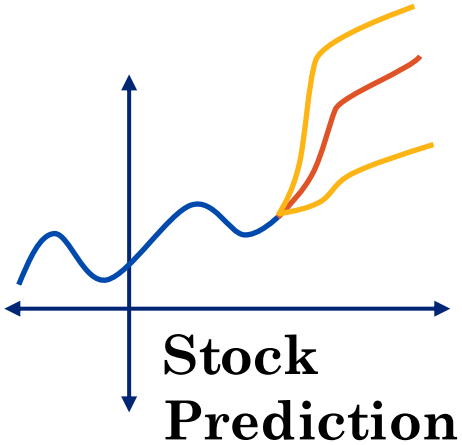
Image Labeling

X: Image
Y: {Hot Dog, …}

ChatGPT

X: Prompt
Y: Next Word

Regression



Stock Prediction

X: History
Y: Next Value

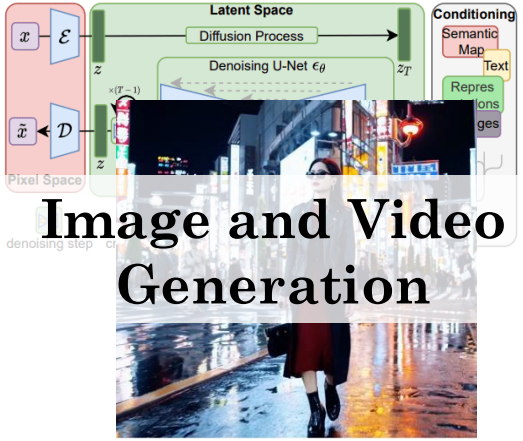


Image and Video Generation

X: Prompt+Noise
Y: Pixel Values

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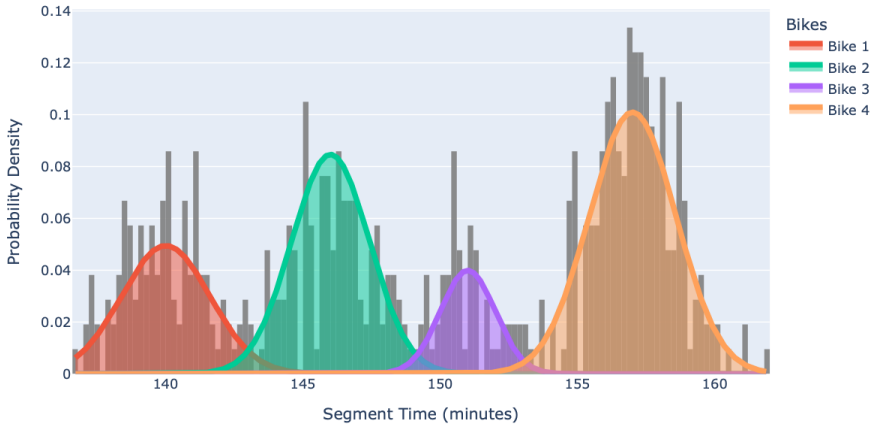
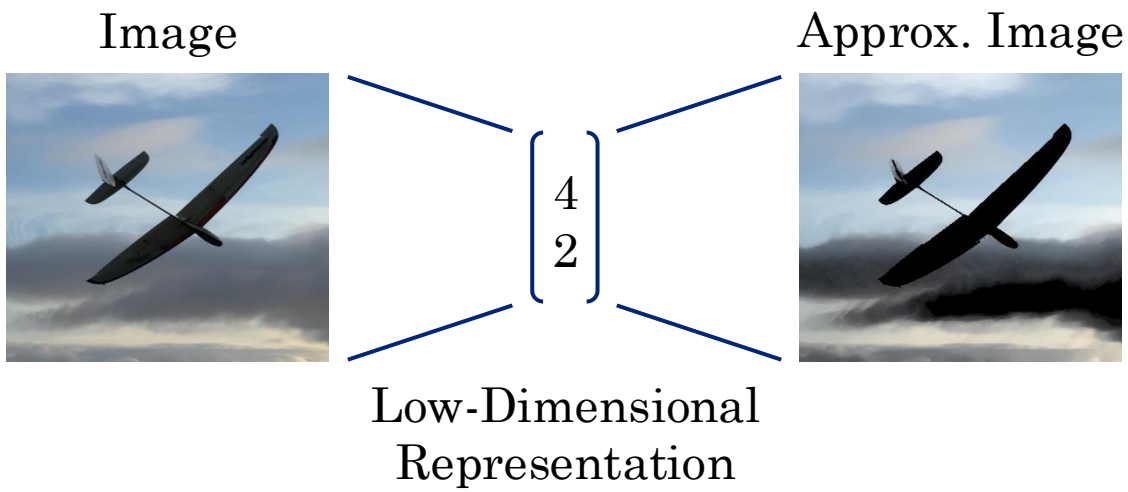
Unsupervised Learning

Trying to **model the data** in the **absence** of **explicit labels**.

Clustering & Density Estimation



Dimensionality Reduction



Used for **visualization** and as a step in **other ML tasks**.

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Reinforcement Learning

Learning from **reward signals** often with complex **multi-step** (discrete or continuous) **action** sequences.

Action: next move,
Reward: Win/Lose

Action: change in joint angles
Reward: Fold quality

ChatGPT

Action: next token
Reward: answer quality

Not covered in this class but a direct extension of topics in this class.

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Types of supervised learning problems (Tabular data)

- Given features about a house predict the sale price.
  (Square feet, number of bedrooms, bathrooms, Zip Code, HasPool, etc)
  https://www.kaggle.com/c/house-prices-advanced-regression-techniques
- Given income, education, number of credit cards etc, predict if an individual will default on their loan
- Instacart Market Basket analysis (Predict which products will an Instacart consumer purchase again)
  https://www.kaggle.com/c/instacart-market-basket-analysis
- Predict sales of Walmart weather sensitive products in 40 locations
  https://www.kaggle.com/c/walmart-recruiting-sales-in-stormy-weather
- Given patient data predict the clinical outcome for a disease.
- Predict next year re-hospitalization using hospital data
  https://www.kaggle.com/c/hhp

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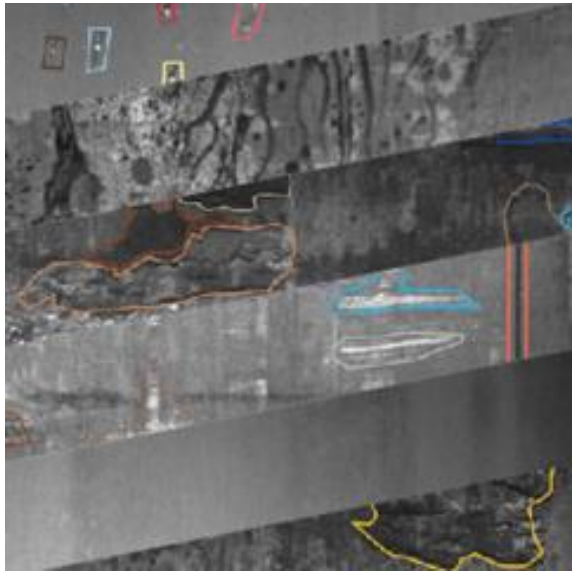
Types of supervised learning problems (Image data / Computer vision)

- Given images of manufactured steel predict there will be defects. https://www.kaggle.com/c/severstal-steel-defect-detection



- Severstal, a leading steel mining and production company wants to use machine learning to automatically predict if manufactured flat sheets have defects.
- Here the features are pixels.

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Types of supervised learning problems (Image data / Computer vision)

- Given images of manufactured steel predict there will be defects.
  (Here features are the pixels of the image, plus usually categorical features)
  https://www.kaggle.com/c/severstal-steel-defect-detection

- Dogs vs Cats: Given an image classify if it is an image of a dog or a cat https://www.kaggle.com/c/dogs-vs-cats-redux-kernels-edition

- Estimate density of wheat heads from outdoor field images https://www.kaggle.com/c/global-wheat-detection

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Types of supervised learning problems (Time series data)

- Predict an upcoming seizure using EEG recordings
- https://www.kaggle.com/c/seizure-prediction

- Covid19 Global forecasting per city
  https://www.kaggle.com/c/covid19-global-forecasting-week-5/discussion/151461

- Speech recognition, Music Genre Classification, Financial time-series forecasting

Types of supervised learning problems (Natural Language processing (NLP))

- Twitter sentiment detection
- https://www.kaggle.com/c/tweet-sentiment-extraction
- (Here the features are the text of the Tweet)

- News Article classification (Politics, Sports, Entertainment)

- Detecting toxic content in online conversations
  https://www.kaggle.com/c/quora-insincere-questions-classification

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A taxonomy for machine learning

- **Supervised learning:**
- I.e. there exists a target variable y we are trying to predict.

Splits to:

1. Binary classification
2. Multi-class classification
3. Regression (y continuous)

- **Unsupervised learning:**
- There is no y

Ok what to do ?

1. Clustering
2. Association analysis (diapers and beer)
3. Anomaly detection
4. Generation of artificial data (create new features that look like natural ones).

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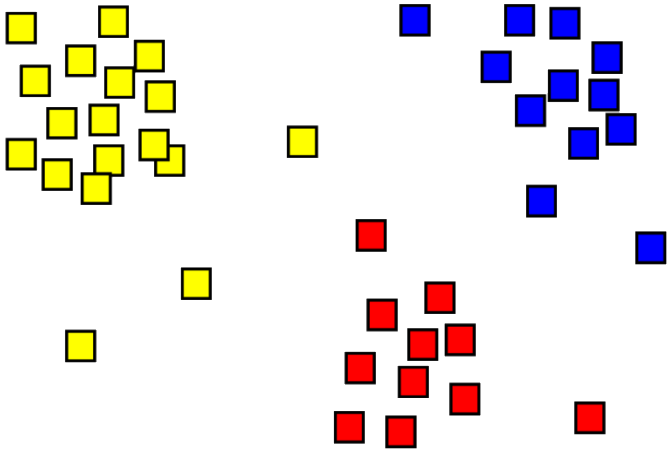
Unsupervised problems: Clustering

- Clustering.
- Consider points in feature space (no labels).
- Assign points in clusters, so that all similar points are assigned together.

- Frequently used algorithms:
- K-means
- DBSCAN



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Unsupervised problems: Dimensionality Reduction

- Dimensionality reduction
- Reduce the number of features, to find a lower dimensional representation.
- Hopefully represents the important directions in dataset

- Frequently used algorithms:
- PCA
- Sparse PCA, Nonnegative PCA, etc
- t-SNE
- Deep-learning representation learning

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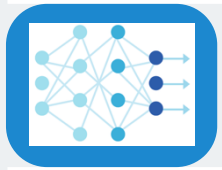
History of Machine Learning

- Introductions
- CS 189/289 Goals and Plans
- What Is Machine Learning?
  - Definition
  - History
- The ML Process (Lifecycle)
- Logistics

History of ML

1950s–60s Early Days	1970s–80s Challenges and Advances	1990s Rise of Statistical ML	2000s Big Data Era	2010s Deep Learning Revolution	Present
Self-learning checkers program (1959) Perceptron (1957)	Decision trees, RL basics, and the rediscovery of NNs	Probabilistic models & statistical learning. Focus on math foundations.	Datasets grew & computation became cheaper Rise of Data Mining & Data Science	Deep learning (2012)	?
					

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History of ML

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				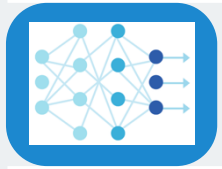	

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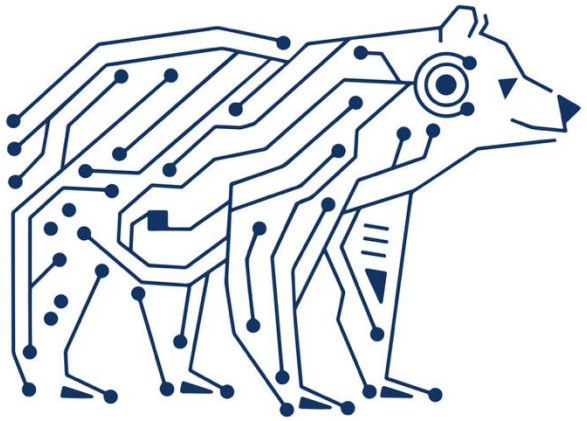
Today – GenAI

What does it mean?	Why is it important?	Will we cover it?
Models that can generate the data.	Unlocking new advanced **general AI abilities**.	**Yes!**

UC Berkeley is at the center of the AI revolution:



Discover BAIR, the world's most advanced academic AI research lab.

Vicuna
AdaGrad
TRPO
NeRF
LLM-as-a-Judge
MMLU

Spark
Caffe
vLLM
Diffusion Models
Chatbot Arena
Gorilla/BFCL

MemGPT
Sleep-time Compute
Test-time Compute
Perplexity.ai
Everybody Dance Now
…

sky riselab UC Berkeley amplab

Get Involved in ***Research!***

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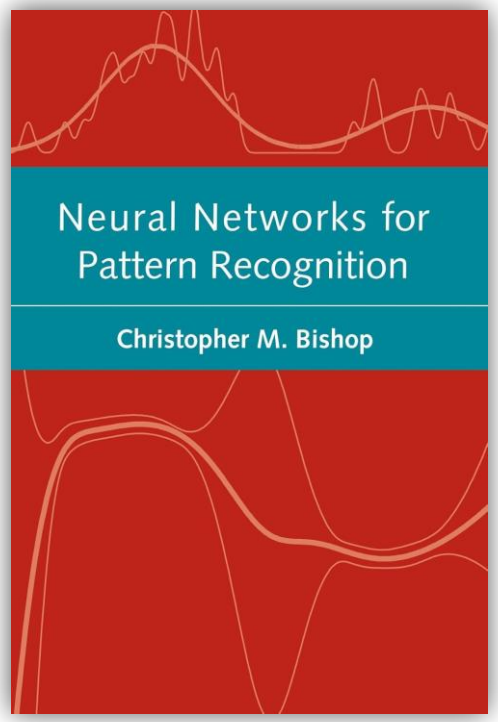
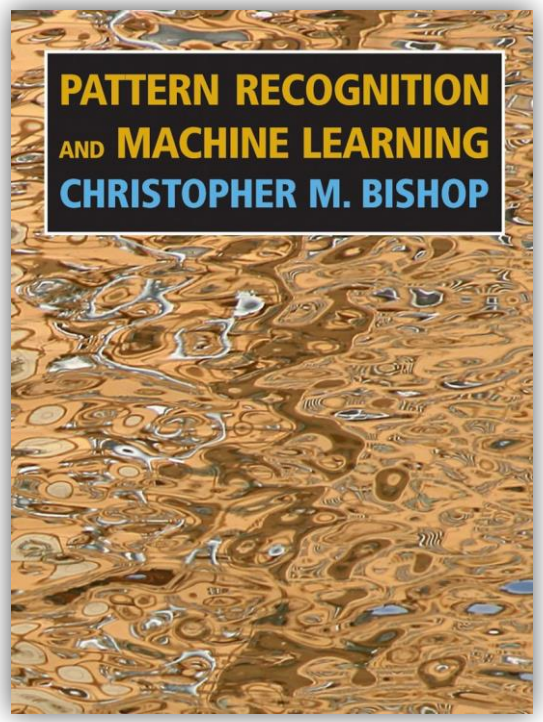
How We teach ML Has *Evolved*

Probability and **Linear Algebra**

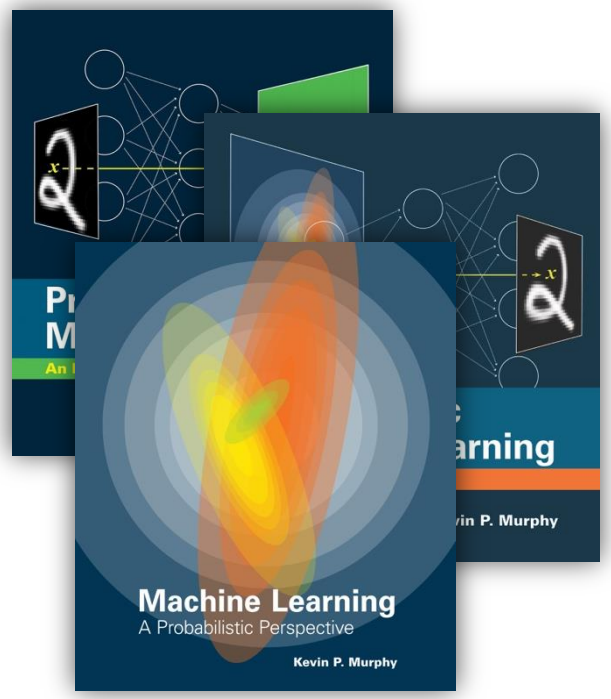
Deep Learning
(NN + Prob + Lin Alg.)



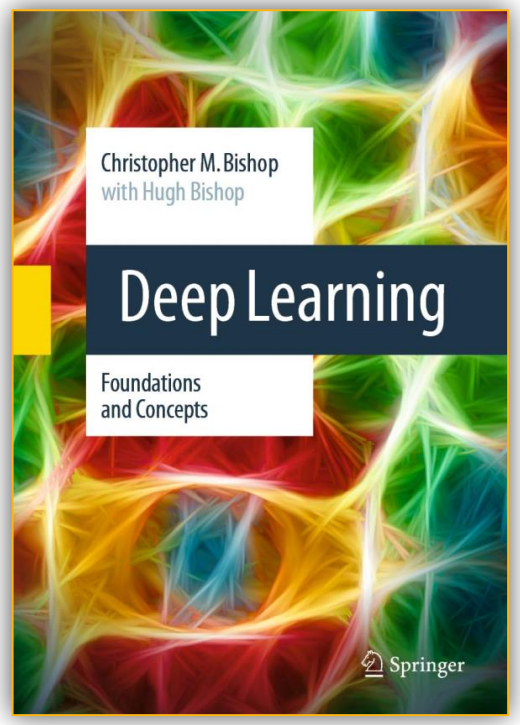
1996



2006



2013-2022



2023

Click on the books to get free PDF versions of all of them!

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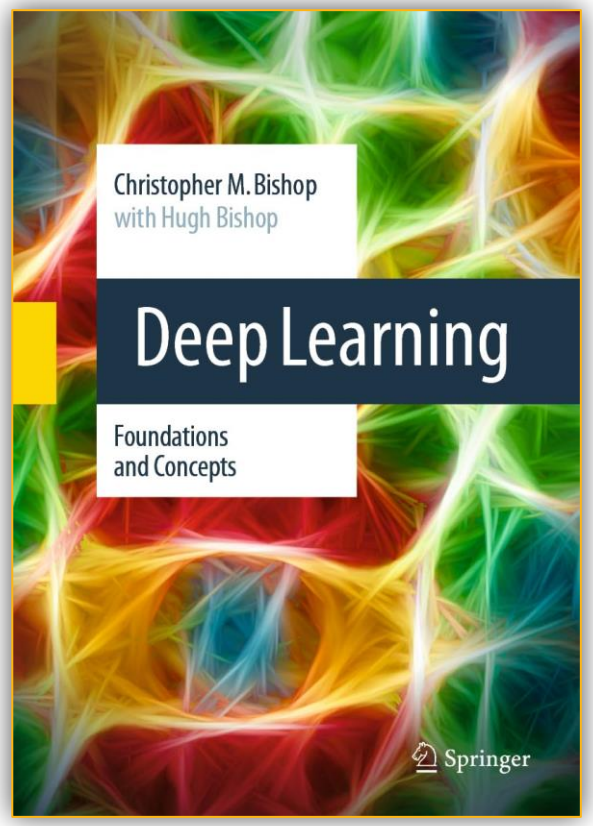
Teaching CS189 w/ Bishop's Latest Book



We will follow the books **notation** and **concepts**

- Book should be a helpful resource

Issues with the book (for this class):

- Heavy emphasis on **probabilistic** framing
- Some material is **out-of-scope for this class**
- Doesn't cover **coding related activities**

Each lecture will have a list of textbook sections that we covered and you're STRONGLY encouraged to read the textbook!

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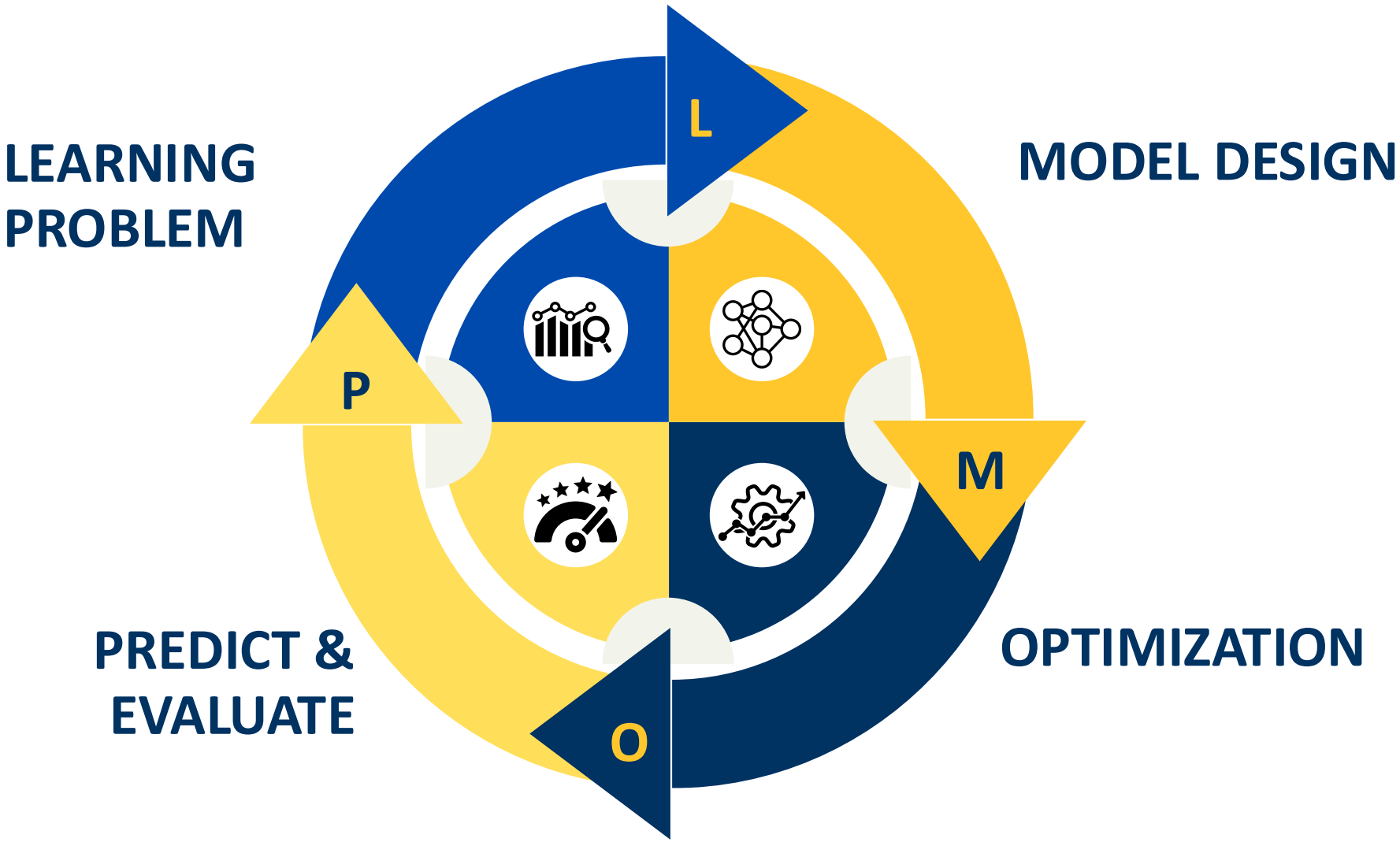
The ML Process (Lifecycle)

- Introductions
- CS 189/289 Goals and Plans
- What Is Machine Learning?
  - Definition
  - History
- The ML Process (Lifecycle)
- Logistics

ML Lifecycle



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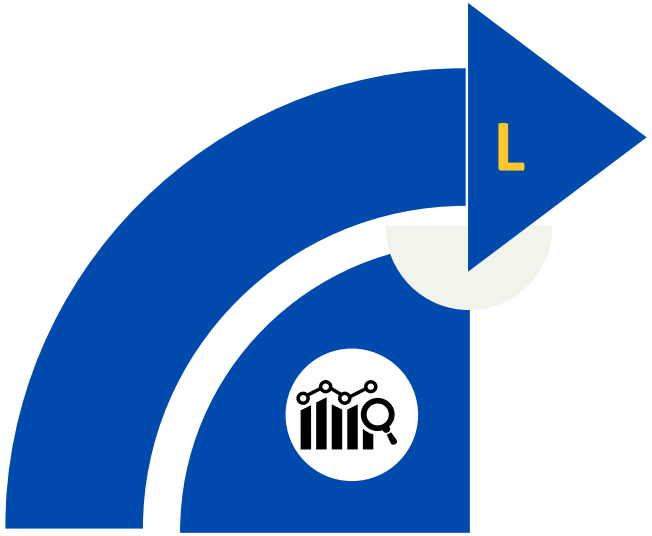
ML Lifecycle

This stage is about framing the real-world question into something a machine learning model can answer.

LEARNING PROBLEM



- **Target:**
  - What do I want to **predict?** What is the machine learning task?
- **Objective:**
  - How would I **evaluate success? What** loss should I use?
- **Data:**
  - What **data** do I have?
  - Data representation? Feature Engineering?
  - Training/Test split

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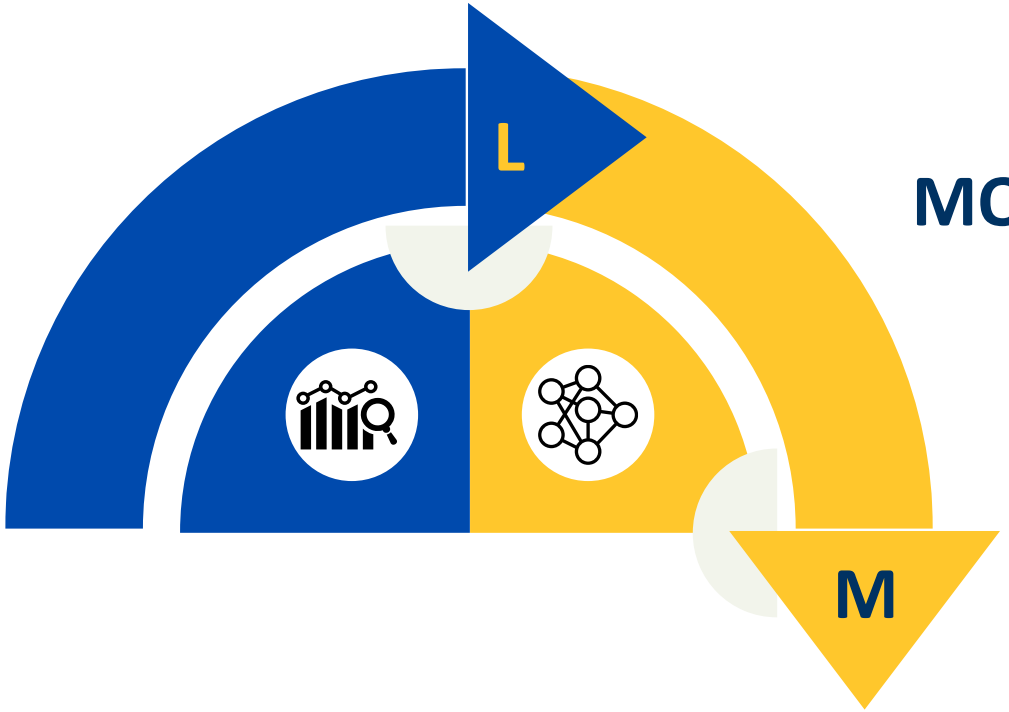
ML Lifecycle

Choose and design an appropriate model.

LEARNING PROBLEM

- Target
- Objective
- Data



MODEL DESIGN

- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

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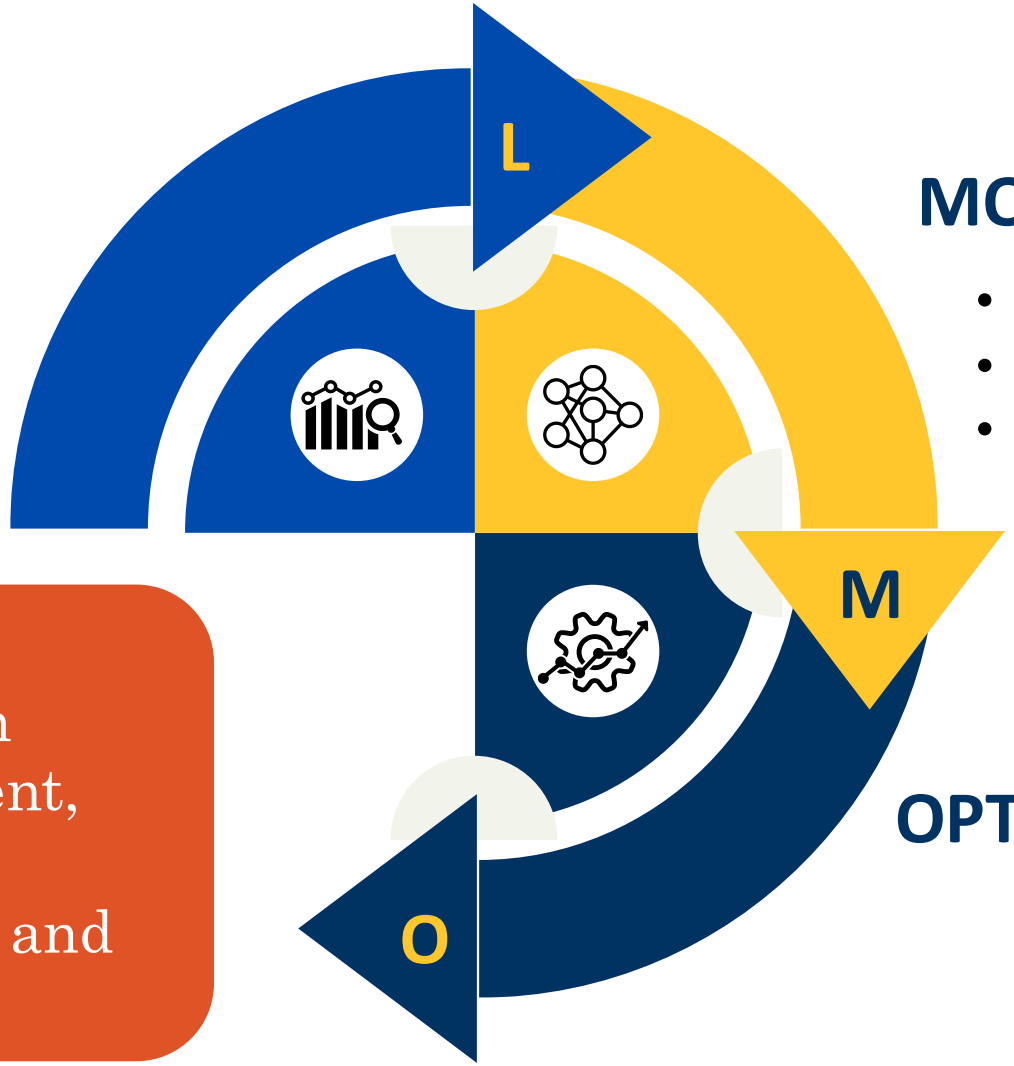
ML Lifecycle

Adjusting the model's parameters to minimize error using optimization algorithms.



LEARNING PROBLEM

- Target
- Objective
- Data

MODEL DESIGN

- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

OPTIMIZATION

- Define a loss
- Choose an optimization method (gradient descent, etc.)
- Manage regularization and overfitting

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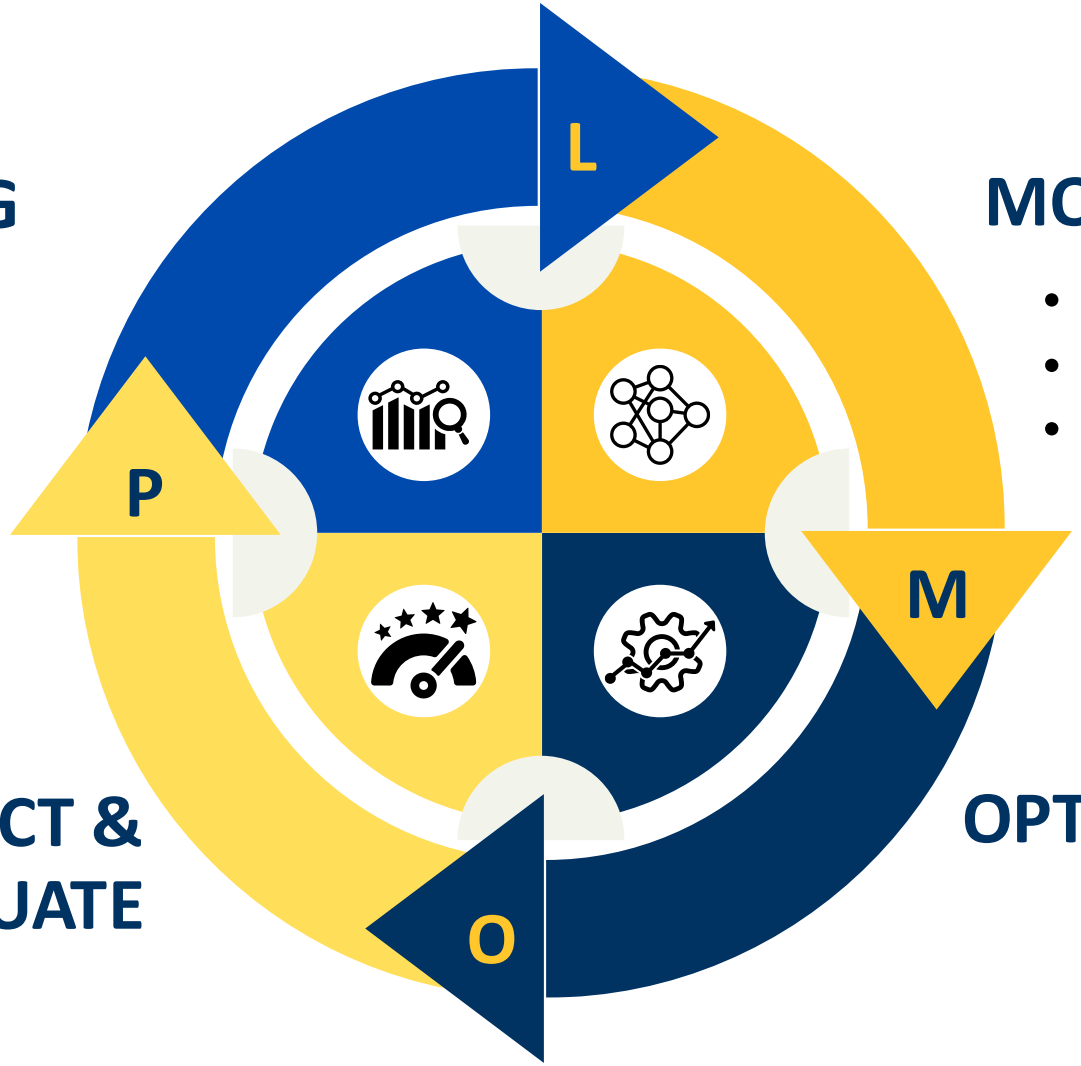
ML Lifecycle

Test how well the model performs.



LEARNING PROBLEM

- Target
- Objective
- Data

MODEL DESIGN

- Model family/Architecture
- Hypothesis space
- Inductive biases / Assumptions

OPTIMIZATION

PREDICT & EVALUATE

Evaluate predictions based on evaluation metrics

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We are going to …

Teach ML “Backwards”

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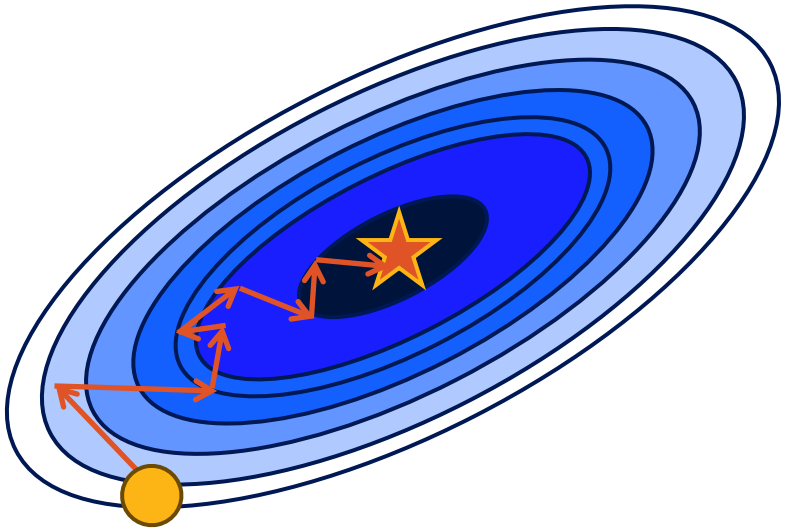
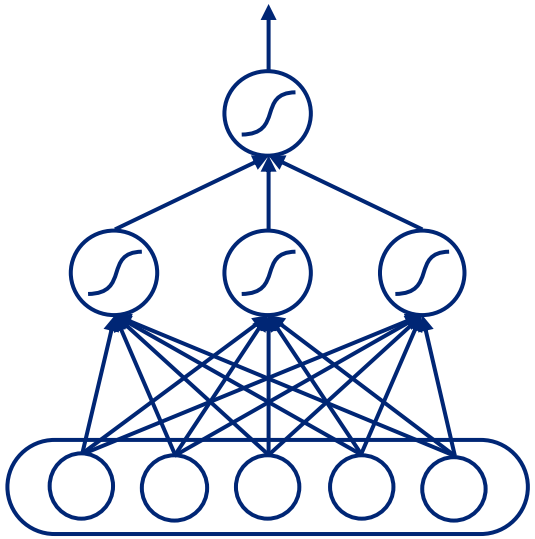
Classic Machine Learning Classes

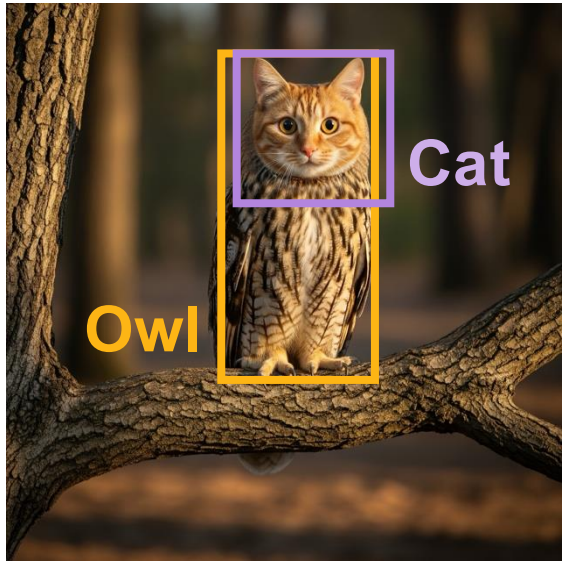
Algorithm



Model



Application



Classic Machine Learning Class

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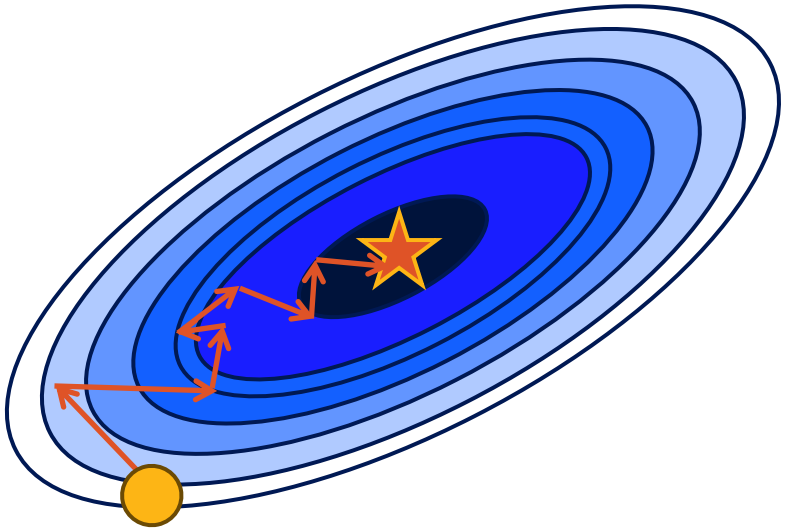
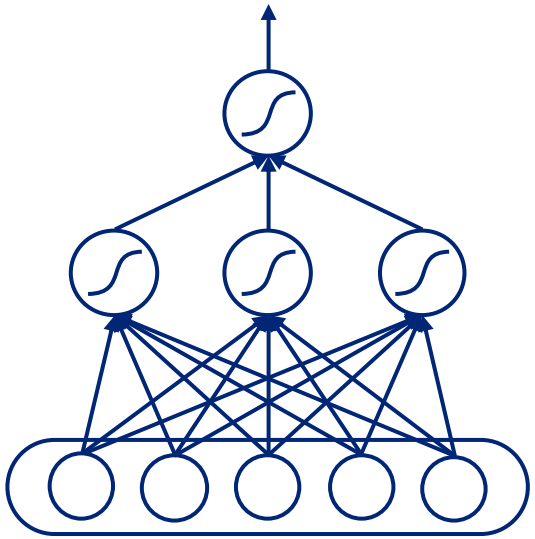
Teaching Machine Learning Differently

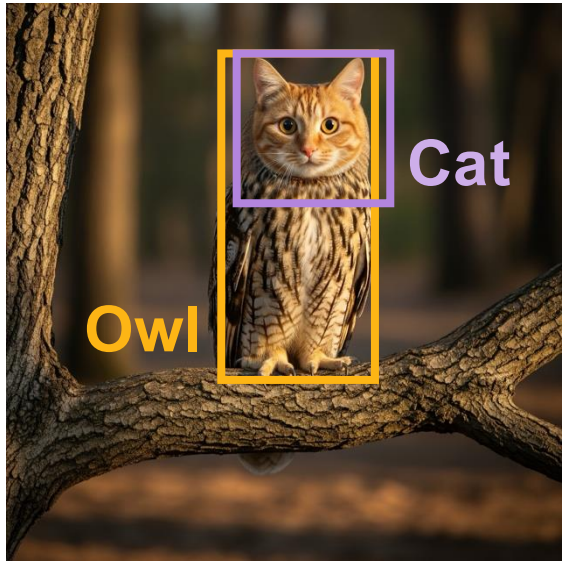
Algorithm



Model



Application



This Machine Learning Class

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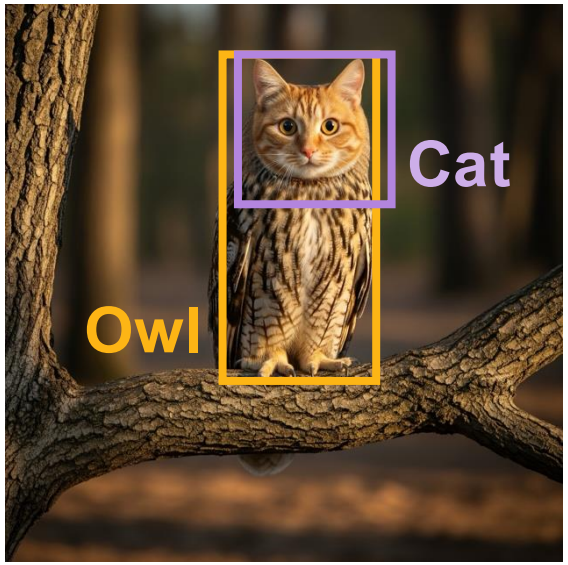
Greater Focus on Application Framing

Application



- When should I use machine learning?
- How do I frame a machine learning problem?
- How do I prepare my data?
- How do I train a model?
- How do I evaluate the model?

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Greater Focus on ML Engineering

You will learn to use **tools for ML**:

- **`pandas`** (https://pandas.pydata.org/) for data manipulation.
- **`Plotly`** (https://plotly.com/) and **`Matplotlib`** (https://matplotlib.org/) for data visualization.
- **Scikit-learn** (https://scikit-learn.org/stable/) for classic machine learning tasks.
- **PyTorch** (https://pytorch.org/) and **HuggingFace** (https://huggingface.co/) for neural network development.
- **Weights-and-Biases** (https://wandb.ai/site/) for experiment management.

We will work in the **Google Colab environment**, but you will also be able to use your own tools if you prefer.

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Logistics

- Introductions
- CS 189/289 Goals and Plans
- What Is Machine Learning?
  - Definition
  - History
- The ML Process (Lifecycle)
- Logistics

Course Map at a Glance

1-2

Introduction and
ML Mechanics

3-6

Supervised Core

k-means/EM;
regression → classification;
GD

7-8

Neural Networks

NN Fundamentals:
backprop
non-linearity
regularization

9

Midterm Week

10-11

Advanced Architectures

CNN
RNN
Transformer
LLM

12-14

Advanced Topics

Generative Models
Autoencoder
Dimensionality Reduction

15-16

Applications

Guest Lecture
More advanced applications

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Assessment Cadence

- **5 Homeworks**, each spans **~3 weeks**, with **two parts** released together:
  - **Part 1 (Warmup)** → introduces tools;
  - **Part 2 (Main)** → deeper application.
- Mix of **written + coding** on Gradescope;
  - Some **public** sanity tests and **hidden** correctness tests.
  - Collaboration = discuss ideas, but **write/code individually**.
  - Open use of GenAI
    - No Vibe Coding/Writing!! You must understand everything you submit!
- **Due 11:59 pm**
  - **No HW drops**.
  - You get **10 total slip days** across the **10 HW deadlines**;
  - **max 4** per HW; slip days **auto-apply** (1 minute late = 1 day).

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CS189

Prerequisites

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Prerequisites

- Official Prerequisite:
  - MATH 53 and MATH 54; and COMPSCI 70 (or equivalent).
- What you need:
  - Good programming skills in Python
    - Implement functions and classes
    - Debug programs
  - Probability and statistics
    - Work with conditional probabilities and Bayes rule and compute expectations
    - Familiarity with uniform, Bernoulli, and Gaussian distributions
  - Linear algebra and Calculus
    - Understand eigenvectors and eigenvalues
    - Compute gradients, apply the chain rule

CS189

Course Staff

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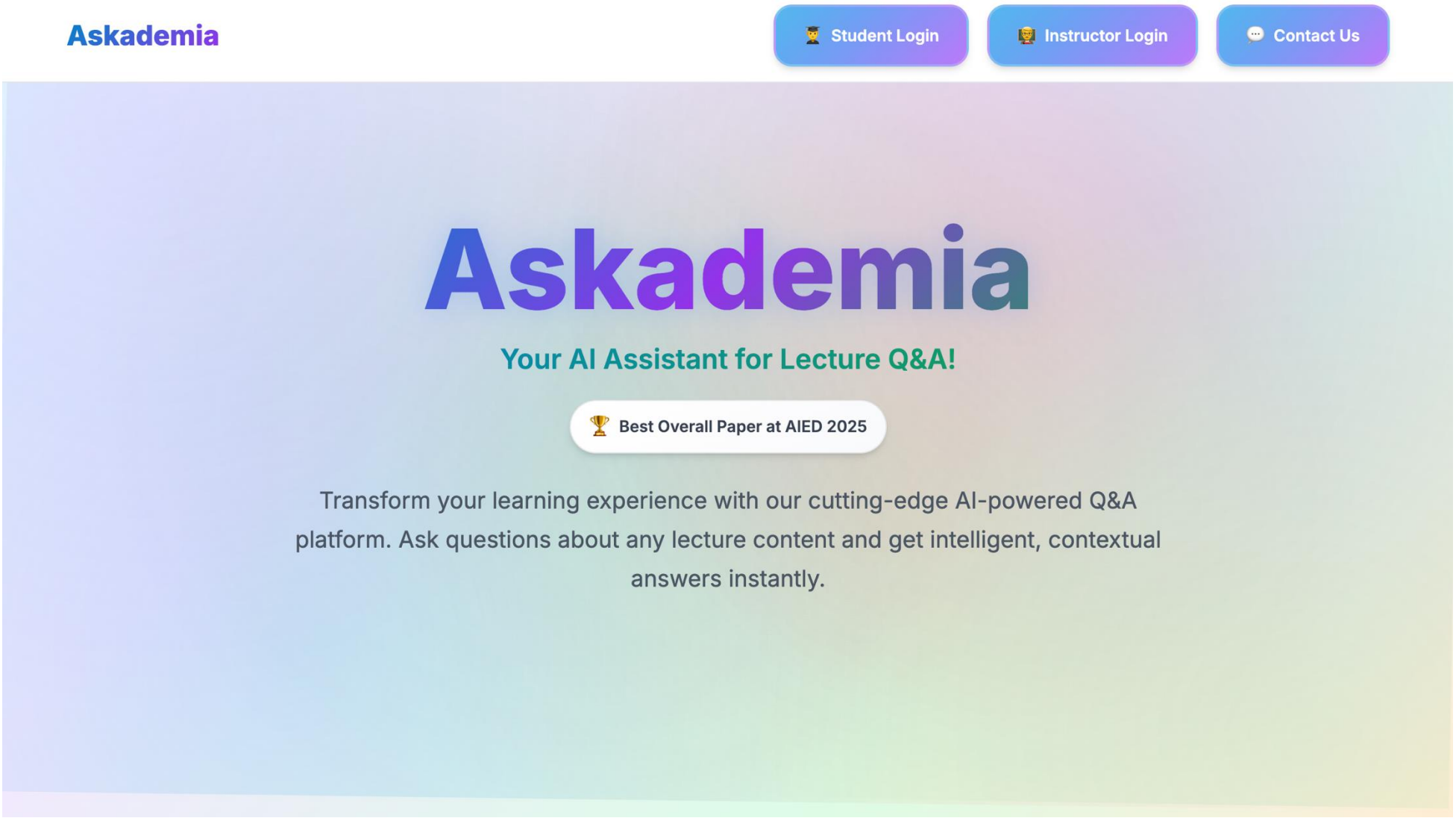
Course Platforms - Askademia



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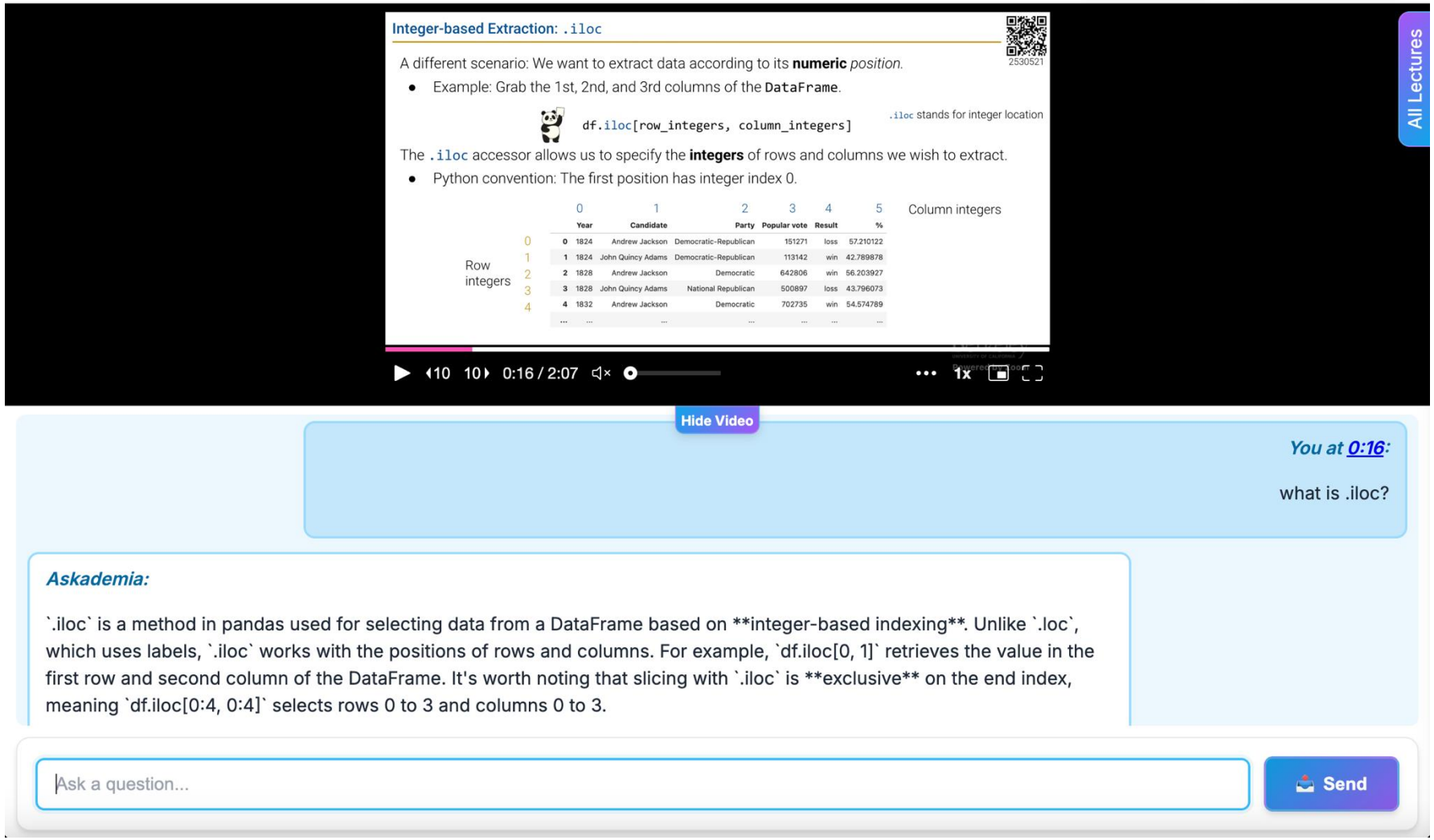
Course Platforms - Askademia



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Lecture 1

Introduction to Machine Learning

Credit: Joseph E. Gonzalez, Narges Norouzi

Reference Book Chapters: Chapter 1 (Section 1.1)